



Faculty Information Systems and
Applied Computer Sciences

M. Sc. International Software Systems Science
University of Bamberg

Master's Thesis

**A Study of Large Language Models' Emotion
Classification in Ambiguous Text with Perceived
Authorship**

Author:

Umer Saeed

Bamberg, September 28, 2026

Professor: **Prof. Dr. Roman Klinger**

Supervisor: **Dr. Sabine Weber**

Abstract

Evaluating emotion recognition in text commonly assumes that each text has a single correct emotion label. Many texts, however, admit several plausible emotional readings, and the one a given reader settles on depends not only on the words but on who they suppose wrote them. Pride, embarrassment, or regret can be conveyed by the sentence “*I can’t believe I actually did that*” depending on who is taken to be the speaker. The thesis asks whether perceived author identity measurably affects the interpretation of emotion, both for human readers and for large language models, and whether the responses of the two are similar.

Three hundred brief self-reports of emotional experience were sampled from the crowd-enVent corpus and stratified into three categories of one hundred texts each: *unambiguous control texts*, *author-independent ambiguous texts*, and *author-relevant ambiguous texts*. Each text was neutrally framed and also presented under two opposite author personas, with the text itself left unchanged. Human judgment was gathered across all three framings. Seven large language models then categorised every text under four framings and six prompt configurations. The outcome is 7,200 model-ensemble distributions and 900 human distributions (one per text and framing condition), constructed using as many as 50,400 model answers and 2,899 human ratings.

The hypotheses are tested in two circumstances: one where a single majority or top emotion is chosen for each condition, and another using the complete **probability distribution over eleven emotions**. When the two are compared to each other, one can see what the single-label method is not able to capture. Among the 300 sampled texts, 19.7% have no emotion commanding a strict majority of readings. Of those, 6.3% still have a unique most-frequent (plurality) label, so a pipeline that reports only the top label can retain it while covering up how divided readers really were; the remaining 13.3% have no single best choice whatsoever, and some arbitrary tie-break must be made. A distribution preserves that disagreement rather than eliminating it. It also allows one to ask not only *whether* an answer changed but *how far* and *in which direction* it changed.

Four hypotheses were tested. As predicted, human judgments shift significantly under author framing in author-relevant ambiguous texts, while unambiguous texts show no detectable shift. There is only one persona contrast on which the author-independent category is significant, and there is no evidence that the two persona effects are different. The model ensemble, however, shifts significantly under both personas. A previously unanalysed **generic_human** control separates out two comparisons with matched models: an unspecified author line alone produces about an 8% move in classifications on both sides, and going further to a named persona is associated with a separately measured 16–19% move (the two distances are not additive elements of a single effect). The predicted category ordering holds in raw shift magnitude. A sensitivity analysis referring each text to its own conditional sampling null, however, shows that at this sample size the ordering cannot be told apart from the dispersion in the baseline votes. Lastly, models move in the same *direction* as human shifts better than chance, though only weakly, and human shifts are substantially larger in *magnitude* than model shifts. This difference in magnitude does not vanish when both sides are standardised to exactly the same number of votes.

Two outcomes are reported as negative or near-negative rather than smoothed out. After testing both human and model shifts, virtually no single per-emotion shift passes multiple-comparison correction (7 of 66, forming a small but directionally coherent pattern on the model side, rather than scattered noise). And the exploratory analysis did not detect a prompt-design effect on the size of the model framing effect. Another finding is methodological in nature, and it reveals most sharply why the representation matters. Run the same statistical test on the chosen label instead of the entire distribution, and it rejects the null in **1** of 6 tests establishing H1 under a reverse-alphabetical tie-break rule, and **0** under the alphabetical rule used throughout this thesis (only these two otherwise-arbitrary rules were tested); the distributional test rejects it in 3 of them regardless. Using a chosen label, this study would have detected a statistically significant human framing effect in at best one of six H1 tests, or none at all under its own convention. More broadly, across all comparisons, two fifths of both human and model comparisons shifted in ways a selected-label analysis would have recorded as no change whatsoever. This value is also convention-dependent: when the label method is credited with tracking the full set of tied top emotions rather than only the alphabetically chosen one, the human figure drops to about a quarter (25.8%), while the model figure stays substantially unchanged (38.7%).

Collectively, these findings support three conclusions: author framing is an actual and measurable factor affecting the interpretation of emotion, language models are sensitive to it, and the resemblance between human and model behaviour is partial and directional rather than close.

Contents

1	Introduction	1
1.1	Motivation	1
1.2	Problem Statement	2
1.3	Research Approach	2
1.4	Contributions	3
1.5	Thesis Structure	3
2	Theoretical Background	5
2.1	Emotion Recognition and Annotator Disagreement	5
2.2	Perceived Authorship and Framing Effects	6
2.3	Points of Intersection	7
2.4	Research Gap and Motivation	7
3	Methodology	9
3.1	Research Questions	9
3.2	Research Hypotheses	9
3.3	Corpus Selection and Dataset Construction	10
3.3.1	The classification rule	11
3.3.2	Drawing the sample	12
3.4	Ambiguity Types and Author Relevance	13
3.5	Author-Framing Manipulation	14
3.6	Realised Experimental Design	15
3.7	Deviations from the Approved Proposal	16
4	Implementation	19
4.1	Prompt Construction and Model Evaluation	19
4.1.1	Stimulus Construction	19
4.1.2	Model Selection	20
4.1.3	Prompt Configurations	21
4.1.4	Response Normalisation and Off-List Answers	21
4.1.5	A Data-Quality Correction Made During Analysis	22
4.2	Human Annotation Study	23
4.3	Emotion Probability Vector Construction	24
4.3.1	Definition	24
4.3.2	Why Distributions Rather Than Labels	24
4.4	Metrics	25
4.4.1	Magnitude: Total Variation Distance	25

4.4.2	Direction: Delta Vectors and Cosine Similarity	25
4.4.3	Dominant Sets and Their Five Outcomes	26
4.4.4	Dispersion Measures	26
4.5	Quality-Control Gate	27
4.6	Statistical Procedures	28
4.6.1	Unit of Analysis	28
4.6.2	Permutation Tests	28
4.6.3	Effect Sizes and Confidence Intervals	29
4.6.4	Multiple Comparisons	29
4.6.5	Referencing Against a Sampling Null	30
4.6.6	Reproducibility	30
5	Results	32
5.1	Disagreement Is Present Before Any Manipulation	32
5.2	Sample Composition	33
5.3	Model Response Validity	33
5.4	Conventional Majority-Label Benchmark	34
5.5	First Pass: H1–H4 with One Selected Emotion	35
5.6	What the Vector Representation Preserves	36
5.7	Two Independent Neutral Measurements Disagree Substantially	37
5.8	H1: Author Framing Shifts Human Judgments	38
5.8.1	Magnitude by Category	38
5.8.2	Permutation Test	38
5.9	H2: Author Framing Shifts Model Classifications	41
5.9.1	Magnitude by Category and Configuration	41
5.9.2	Paired Permutation Test	42
5.9.3	What This Test Does and Does Not Establish	43
5.10	Is It the Persona, or Merely Having an Author?	44
5.11	H3: Framing Effects and Ambiguity Type	45
5.11.1	Raw Contrasts	45
5.11.2	Referencing Against Each Text’s Own Sampling Null	45
5.12	H4: Human–Model Alignment	47
5.12.1	Magnitude	47
5.12.2	Is the Magnitude Gap an Artefact of Vote Counts?	48
5.12.3	Direction	50
5.13	Where the Probability Mass Moved	51
5.14	A Traceable Strong-Shift and Stable-Condition Example	53
5.15	Does Framing Concentrate or Disperse Interpretation?	54
5.16	How Often Does Framing Change Nothing?	56
5.17	The “Unchanged” Category Conceals Movement	56
5.18	Prompt Structure Has No Measurable Effect	57
5.19	Supplementary Check: Human Rating Agreement	58
5.20	Summary of Hypothesis Outcomes	59

6	Discussion	61
6.1	Interpretation with Respect to the Research Questions	61
6.1.1	RQ1: Does Perceived Authorship Influence Human Emotion Judgments?	61
6.1.2	RQ2: Does Perceived Authorship Influence LLM Classifications?	62
6.1.3	RQ3: Do LLMs Resemble Humans in Direction and Magnitude?	63
6.2	Cross-Cutting Findings	64
6.2.1	What Aggregation Costs	64
6.2.2	Small Samples Make TV Deceptive	65
6.3	Limitations	65
6.4	Future Directions	67
7	Conclusion	69
7.1	Summary of Findings	69
7.2	Contribution	70
7.3	Closing Remarks	71
	Bibliography	73

List of Tables

3.1	Realised experimental design.	16
5.1	Vote-pattern distribution across the three judgment sources.	33
5.2	Excluded (off-list or missing) answers by model, out of 7,200 judgments each.	34
5.3	Neutral model performance against the original readers' strict majority, averaged over six prompt configurations.	35
5.4	First-pass conclusions using selected emotion labels only. The full calculations are in the accompanying old-method CSVs.	35
5.5	What a selected-label analysis would record. These three outcomes are exhaustive and mutually exclusive; only the middle row is invisible to a label-based pipeline, which records it as unchanged despite the distribution having moved.	36
5.6	Mean TV distance between the original corpus neutral round and this study's neutral round.	37
5.7	H1 permutation test results. Holm-corrected across six tests (3 categories $\times$ 2 persona sides).	40
5.8	H2 paired sign-flip permutation test, Holm-corrected across two persona sides.	42
5.9	Decomposition of the model framing effect using the generic-human control. Each row uses only the models that answered validly under all three of the neutral, generic-human and persona conditions for that side (a three-way intersection, stricter than matching models pairwise), so the two component distances and the confounded distance are computed over the same model set.	44
5.10	H3 pairwise contrasts on raw TV, Holm-corrected across the three contrasts.	45
5.11	Observed, expected-under-null, and excess TV by category (human, text-level).	46

5.12	H3 contrasts on raw TV versus null-referenced excess TV.	46
5.13	Magnitude gap before and after standardising both sides to three votes. .	49
5.14	Per-emotion shifts significant under BH-FDR at $q < 0.05$ (of 66 tests). . .	51
5.15	Entropy change under framing (text-level, BH-FDR corrected across six tests).	54
5.16	Dominant-set outcomes. These set relationships differ from changes in an alphabetically selected single winner.	56
5.17	Human Fleiss' κ by sampled category and framing; included cells in paren- theses.	58
5.18	Summary of hypothesis outcomes.	59

List of Figures

5.1	Human persona-effect magnitude by text category (text-level, one observation per text).	39
5.2	Permutation null distributions for the human persona effect, pooled across all texts, with the observed mean marked.	40
5.3	Model ensemble persona-effect magnitude by text category (text-level, averaged across the six prompt configurations).	42
5.4	Model persona effect by exact prompt configuration. Each bar is a separately computed, never-pooled comparison.	43
5.5	Mean human versus mean model shift magnitude for each of 300 texts, coloured by category.	48
5.6	Distribution of per-comparison directional cosine between human and model delta vectors, one point per (text, side, prompt-configuration) comparison, with the observed mean and the chance-pairing null marked. . . .	50
5.7	Mean signed change in probability mass per emotion and category (human, text-level, percentage points).	52
5.8	Rates at which emotions appear or disappear between conditions, by source and category.	53
5.9	Mean entropy change from neutral to persona framing, by source and category. Positive values indicate greater dispersion.	55
5.10	Dominant-emotion transition matrices for human and model judgments, restricted to comparisons where both sides have a unique dominant emotion. .	57
5.11	Per-text shift magnitude across the six prompt configurations.	58
5.12	Overview of hypothesis outcomes across the analysis.	60

List of Abbreviations

ρ	Spearman’s rank correlation coefficient
κ	Fleiss’ kappa, the inter-annotator agreement statistic used in Section 5.19
AI	Artificial Intelligence
API	Application Programming Interface
BH-FDR	Benjamini–Hochberg False Discovery Rate correction; used for exploratory test families
CI	Confidence Interval
Clustered bootstrap	Bootstrap resampling texts rather than rows, so that repeated observations of the same text are not treated as independent
Cosine similarity	Angle-based similarity between two delta vectors, used to compare shift direction
crowd-enVent	Corpus of self-reported emotional events with author and reader annotations (Troiano et al., 2023)
Delta vector	Signed difference between a persona-condition vector and its neutral-condition vector
Dominant set	The set of emotions attaining the maximum probability in a vector; a set rather than a single label because ties are frequent
Excess TV	Observed TV minus the TV expected for that text under its own conditional sampling null
Few-shot	Prompt variant providing several worked examples
Generic human	Framing condition attributing the text to an unspecified human author
H1–H4	The four research hypotheses stated in Section 3.2

Holm	Holm correction for multiple comparisons, controlling the family-wise error rate; used for confirmatory tests
LLM	Large Language Model
Neutral	Framing condition in which no author information is supplied
Numbered inline	/ The two label-presentation formats: emotions as a numbered list, or embedded in the question text
One-shot	Prompt variant providing a single worked example
Paired sign-flip	Permutation null in which each model’s own two answers are swapped with probability one half, holding model identity fixed
Permutation test	Significance test comparing an observed statistic against a null built by relabelling the observed data
Persona A / B	The two contrasting author personas assigned to each text
pp	Percentage points
Pseudoreplication	Treating repeated, correlated observations of the same unit as independent observations
QC	Quality Control
RQ	Research Question
Support size	Number of emotions receiving non-zero probability mass
TV	Total Variation distance, $\frac{1}{2} \sum_i p_i - q_i $; the share of probability mass separating two distributions
Zero-shot	Prompt variant providing no worked example

1 Introduction

Emotion recognition in text tries to identify the affective state expressed or implied in written language. Sentiment analysis typically reduces to positive and negative polarity; emotion classification instead assigns finer-grained categories such as joy, anger, fear, shame, pride or sadness. The task is hard because emotional meaning is rarely stated outright. It arises from the interaction of linguistic content, situational context, and the assumptions a reader makes about the person speaking.

The major challenge is ambiguity. Some texts allow a variety of reasonable emotional readings, due to underspecified context, an emotionally complicated situation, or the relative position of writer and reader. Even short, seemingly simple statements can be read in competing ways. Consider:

I can't believe I actually did that.

Without further context, this could be a statement of pride, embarrassment, regret, excitement, or disbelief. Background assumptions tend to resolve such ambiguity for the reader, and one of the most accessible is an assumption about *who* is speaking. Attributed to someone running their first marathon, the sentence invites one emotional reading; attributed to someone recounting a lapse of judgment, it invites a different one, yet the words are the same.

1.1 Motivation

The thesis has two motivations.

The first is computational. Large language models are now widely used for text classification, emotion recognition included, and are known to be sensitive to prompt framing and contextual cues. If minimal author information systematically changes a model's emotional interpretation, then an emotion classifier's output is shaped by the social con-

text supplied in the prompt, not by text content alone. That bears directly on how such systems are evaluated and deployed.

The second is cognitive and social: do models reflect human interpretive behaviour when dealing with ambiguous emotional content? Humans demonstrably use assumptions about a speaker to disambiguate. Whether models do the same, and to what extent and in what direction, is an empirical question. It has not been directly investigated by manipulating perceived authorship while holding the text constant.

1.2 Problem Statement

Current research on emotion recognition largely treats disagreement among annotators as noise to be removed. Datasets often include a single aggregated label for each text, most frequently by majority vote, and models are trained and evaluated against that label. Built into this practice is an assumption that fails for a large portion of ambiguous texts: that there is one correct emotional reading and that annotators who do not stay with that reading are simply in error.

The problem addressed here has two parts. The first is whether perceived author identity has a systematic effect on emotion interpretation, in both humans and models, or only affects it incidentally. The second is whether the normal practice of collapsing judgments to a single label can detect such an influence at all. Suppose author framing causes a shift in the *distribution* of plausible readings without necessarily changing which reading is most popular. An analysis based on a majority label will then record no change where an actual change occurred.

1.3 Research Approach

The thesis manipulates perceived authorship experimentally and keeps the textual content fixed. Three hundred texts from the crowd-enVent corpus [Troiano, Oberländer, and Klinger 2023] were stratified into three groups: unambiguous control texts, author-independent ambiguous texts, and author-relevant ambiguous texts. Each text was framed neutrally and under two contrasting author personas. Human annotations were collected for all framings, and seven large language models labelled every text across four framings and six prompt settings.

What distinguishes this work is its choice of representation. All judgments, human and model, are modelled as full probability distributions over the label set of eleven emotions rather than as a single label. Framing effects are then shifts in that distribution: their *magnitude* is the total variation distance, their *direction* the signed change in probability mass across emotions. Using this, one can distinguish a text that was only slightly reinterpreted from one that was substantially reinterpreted but happened to retain the same most-popular answer.

1.4 Contributions

The contributions of this thesis are as follows.

- An experimental design that dissociates the influence of perceived authorship on emotion interpretation by changing author framing while holding text fixed, implemented in parallel on human annotators and seven large language models.
- A distributional analysis framework in which each judgment is an eleven-dimensional probability vector, with the magnitude and direction of framing-induced shift measured separately and never mixed together.
- Empirical evidence that author framing measurably shifts both human and model emotion judgments, together with a control condition that dissociates the effect of the persona’s content from the mere presence of an author attribution.
- A direct measure of what majority-label analysis gives up: the fraction of judgments that shifted but would have been recorded as unchanged.
- An explicit account of what these data can and cannot establish, including a sensitivity analysis showing that, at this sample size, the predicted ordering across ambiguity categories cannot be separated from baseline disagreement.

1.5 Thesis Structure

Chapter 2 examines research on emotion recognition and annotator disagreement, and on framing effects arising from perceived authorship, and identifies the gap this thesis addresses. Chapter 3 describes the corpus, how texts were categorised by ambiguity type, and the author-framing manipulation, together with the research questions and

hypotheses. The entire pipeline is recorded in chapter 4: prompt construction, model evaluation, the human annotation study, the generation of emotion probability vectors, the metrics derived from them, and the quality-control and statistical procedures. Chapter 5 is a hypothesis-by-hypothesis report of the results. These results are explained in relation to the research questions in chapter 6, and the limitations of the design are set out, along with recommendations for future research. Chapter 7 concludes.

2 Theoretical Background

This thesis brings together two research strands. One is emotion recognition and the status of annotator disagreement within it; the other is the framing effect of perceived authorship. Section 2.3 shows where the two meet, and Section 2.4 states the gap this thesis addresses.

2.1 Emotion Recognition and Annotator Disagreement

Emotion recognition assigns fine-grained affective labels to text, whereas sentiment analysis has typically been reduced to polarity. Popular resources such as GoEmotions [Demszky et al. 2020] provide large annotated corpora over rich emotion taxonomies, and transformer classifiers trained on them achieve good benchmark performance [Acheampong, Nunoo-Mensah, and Chen 2021].

Beneath these benchmarks lies a common assumption. Each text is labelled by multiple annotators, their judgments are aggregated (usually by majority vote), and the aggregate is the gold standard against which models are trained and scored [Basile 2020]. On this view, annotators are equivalent estimators of one underlying truth, and any difference from the majority is a mistake.

A large literature argues that this assumption does not hold for subjective tasks. Pavlick and Kwiatkowski 2019 show that disagreement in natural language inference is often *inherent* rather than erroneous: annotators differ genuinely and reproducibly. Uma et al. 2021 survey learning approaches that model disagreement instead of eliminating it. Aroyo and Welty 2015 argue directly against the single-ground-truth perspective, and Davani et al. 2022 show that models treating annotators individually can outperform those trained on aggregated labels.

When disagreement is systematic, aggregation is not neutral. Annotator judgments vary with demographic background [Prabhakaran, Davani, and Díaz 2021], and in hate

speech and offensive language annotation the variation follows social group membership [Sap et al. 2019; Waseem 2016]. Nor is it only demographic. Geva et al. 2019 show that, for datasets whose instances were themselves written by crowdworkers, a model can learn to identify the most prolific annotators from the text they wrote, and that the model generalises poorly to examples written by annotators unseen during training. The identity of the annotator itself, that is, carries systematic, non-interchangeable signal. A majority label therefore encodes the dominant reading and suppresses minority ones [Davani, Díaz, and Prabhakaran 2022].

A particularly clear example is emotion. According to Acheampong et al. 2021, emotional meaning is often implicit and context-dependent, leaving room for competing but defensible readings. Chou and Lee 2019 argue that each rating contains information, so disposing of non-majority ratings would mean disposing of evidence.

2.2 Perceived Authorship and Framing Effects

The second strand concerns the role that assumptions about a speaker play in interpreting what is said. The same text can be read differently depending on who is believed to have written it.

Henestrosa et al. 2023 found that, for a science journalism article held constant in content, authorship cues had an impact on readers’ perceptions of the *author*, and that their experiments did not reveal differences in the article’s perceived credibility or trustworthiness between AI and human authorship. The manipulation is subtle (an attribution line, not a change in content), but it shifts some judgments and not others. The present thesis adopts the same design: hold the text fixed, vary only the attributed author, and quantify the difference, without determining in advance what judgments will vary and which will not.

Contextual framing sensitivity is well documented for language models. Santurkar et al. 2023 demonstrate that model outputs on opinion questions conform to some demographic groups more than others, and that this shifts when the model is prompted to adopt a persona. Zhang et al. 2018 show that conditioning dialogue agents on persona profiles systematically changes their responses. Both results establish that persona information in a prompt is not neutral.

Work applying models to ambiguous emotional content has begun to treat ambiguity as a property in its own right. Hong et al. 2025 propose ambiguity-aware emotion recognition with large language models, arguing that a single label is not always an appropriate output target. Röttger et al. 2022 distinguish annotation paradigms aimed at a single, correct answer from those aimed at intentionally representing variation.

2.3 Points of Intersection

Neither strand has fully explored the point at which the two meet.

The literature on disagreement shows that variation in subjective annotation is systematic and worth preserving, but it tends to locate that variation in the *annotator*: their background, their sensitivity, their interpretive style. The variable is who is reading.

The framing literature shows that assumptions about the *author* shift interpretation. It usually measures the shift on a single outcome variable such as perceived quality, however, and typically in humans or in models, not in both under one design.

Together, they raise a question that neither strand asks directly. Suppose disagreement in emotion annotation is partly driven by differing assumptions about the author, which readers make silently when context is underspecified. Supplying that context explicitly should then shift the *distribution* of emotional readings in a measurable and predictable way. If models are sensitive to persona framing, they should show a similar shift that can be compared against the human shift on the same texts.

2.4 Research Gap and Motivation

The review above points to three gaps.

Perceived authorship is not manipulated as an experimental variable in emotion recognition. Existing emotion corpora record disagreement but do not vary author attribution while holding the text unchanged. Disagreement is observed rather than induced, so its link to assumptions about the author cannot be isolated.

Aggregation destroys the quantity of interest. If author framing redistributes probability mass among plausible emotions without necessarily changing which emotion is most popular, a majority-label analysis is structurally insensitive to it. The effect

can only be seen when judgments are expressed as distributions. This thesis measures directly how much aggregation loses (Section 5.6).

Human and model framing effects are rarely measured under one design.

Claims that models do or do not resemble human interpretation are often based on comparison with a majority-derived gold label. That tells us whether the model’s single answer matches the human’s. It does not tell us whether the model is *responsive to context* as human beings are. Measuring the change on both sides, under the same manipulation, on the same texts, and comparing direction and magnitude separately, is a different and narrower question.

This thesis takes up all three. It manipulates perceived authorship experimentally, represents every judgment as a full emotion distribution, and applies the identical manipulation to human annotators and to seven large language models, so that their responses can be compared directly.

3 Methodology

The research questions and hypotheses are introduced first, followed by the corpus, the typology of ambiguity types in texts, and the author-framing manipulation. The chapter ends with the experimental design as it was actually carried out and an account of how that design differs from what was originally proposed.

3.1 Research Questions

The thesis addresses three research questions.

- **RQ1 (Framing Effect in Humans).** Does perceived author identity significantly influence human emotion judgments of ambiguous texts?
- **RQ2 (Framing Effect in LLMs).** Does perceived author identity significantly influence emotion classifications produced by large language models?
- **RQ3 (Human–Model Alignment).** Are the framing effects of LLMs similar in direction and magnitude to those of human annotators?

3.2 Research Hypotheses

These questions are operationalised in four hypotheses.

- **H1 (Human Framing Hypothesis).** Human emotion judgments will differ significantly across author-framing conditions for ambiguous texts, but not (or substantially less) for unambiguous control texts.
- **H2 (Model Framing Hypothesis).** LLM emotion classifications will shift across author-framing conditions, reflecting sensitivity to minimal author cues.

- **H3 (Ambiguity Sensitivity Hypothesis).** Framing effects will be stronger for author-relevant ambiguous texts than for author-independent ambiguous texts.
- **H4 (Alignment Hypothesis).** The direction of framing-induced shifts in LLM classifications will partially align with human judgment shifts, but the magnitude of these shifts will differ.

The scoring of the evidence for H4 is affected by two aspects of its wording. First, H4 predicts *partial* directional alignment, so a positive but weak directional correlation is counted *for* the hypothesis, rather than against it. Second, H4 predicts that magnitudes will *differ*. Human and model shift sizes that turned out similar would thus be a surprise with respect to the hypothesis as stated, not an affirmation of it. Chapter 5 scores the two components individually.

The approved proposal contains no numbered hypothesis about prompt structure. Prompt-configuration effects are thus treated as an exploratory question, not as a test of a formal hypothesis.

3.3 Corpus Selection and Dataset Construction

The source of the data is the crowd-enVent corpus [Troiano, Oberländer, and Klinger 2023], which comprises short self-authored descriptions of emotional events. Every text has two layers of annotation: one label provided by the author of the text, indicating the emotion they were trying to express, and labels provided by five independent readers who did not see the author’s label. These two layers make the corpus suitable for this study: disagreement can be assessed both between author and readers and among the readers themselves.

The corpus uses eleven categories of emotion, consisting of *anger*, *disgust*, *fear*, *guilt*, *joy*, *pride*, *relief*, *sadness*, *shame*, *surprise* and *trust*. All eleven are kept. The set is small enough for human annotators and models to work with, and deleting any category could suppress exactly the disagreement patterns the study aims to observe. This fixed, alphabetically ordered set of eleven is the dimension set for every probability vector built in Chapter 4.

To operationalise ambiguity, two quantities were calculated per text. **Author–reader agreement** asks whether the reader majority recovers the emotion the author claims to have intended; a discrepancy means the text failed to convey what its author intended.

Reader consensus (*majority share*) is the fraction of the five readers who selected the majority emotion. Since there are exactly five readers, this measure only takes the values 0.2, 0.4, 0.6, 0.8 and 1.0, a coarseness that directly constrains where a threshold can sit.

Reader consensus does not cluster at the top of that scale. In the candidate corpus, a majority share of 0.6 (three readers of five) occurs most frequently. Disagreement is the typical case and not an edge case, and this is the empirical basis for the distributional representation adopted in Section 4.3.

3.3.1 The classification rule

Three decisions are applied in a fixed order. Their thresholds and ordering are encoded in `scripts/classify_ambiguity_pools.py`. **The persona-flip input for each text (Step 2 below) was generated by an LLM judge, not by human pilot annotators:** the judge was shown the same text under two candidate personas and asked for the emotion it read into each, and the two judgments were compared directly. This LLM-judging step is described in the dataset-construction notebook, `complete_analysis.ipynb`; the specific model used as the judge is not recorded in the materials retained for this thesis and could not be reconstructed at the time of writing. The full derivation, including a threshold sensitivity analysis, is documented in `00_corpus_and_sampling.ipynb`.

Step 0: exclude off-label texts. Texts whose author emotion or reader majority is *boredom* or *no-emotion* are dropped, since these fall outside the eleven-emotion set. Forced into the taxonomy, they produce persona-to-persona differences that are a labelling artefact, not a real interpretive shift. The step removes 215 of the 1,200 candidates and leaves 985 eligible.

Step 1: threshold first. A text is **unambiguous** if majority share ≥ 0.8 *and* the readers recover the author’s own label. This is decided purely from the original corpus’s reader data.

Step 2: flip test, among the remainder only. Of the texts not classified as unambiguous, those whose two candidate personas yield different judged emotions from the LLM judge are **author-relevant ambiguous**; the rest are **author-independent ambiguous**.

The ordering is deliberate. A persona flip does *not* override a strong, author-matching reader consensus: a text the readers already agree on remains unambiguous even when the LLM judge is moved by a persona flip. With the order reversed, the flip test could shift high-consensus texts into the author-relevant category, and that category’s definition would then become circular with respect to H3.

A filtering inconsistency at this step. Step 0 filters out texts whose author emotion or reader majority does not lie within the eleven-emotion set, on the grounds that forcing an off-label judgment into the taxonomy is an artefact of labelling and not a true interpretive shift. The same criterion was not applied to the LLM judge’s own answers at Step 2: of the 100 texts classified as author-relevant, 24 have the judge answering *no-emotion* for at least one persona, precisely the type of off-label judgment Step 0 treats as unreliable elsewhere. These texts were not re-screened against that criterion, so the author-relevant category is filtered less consistently than the unambiguous category.

With a five-valued measure, only three threshold settings are defensible, and 0.8 is one of them. At 0.6 the unambiguous pool would take in texts on which only three of five readers agreed, weakening the control category the design relies on; at 1.0 it would admit only unanimous texts. Every one of the three leaves more than 100 candidates per pool, so availability does not force 0.8; it was chosen as the stricter of the two middle options.

3.3.2 Drawing the sample

The sample was not a single fresh draw and should not be described as one. It was produced by **reconciliation** against an existing sample: the current sample is kept, every text in it is reclassified under the rule above, any pool of more than 100 texts is pruned back to 100 using the fixed seed (42), and shortfalls are filled by drawing from eligible candidates not already used. Three texts set aside as one-shot and few-shot prompt examples are excluded throughout, so that no sampled text can appear inside its own prompt. This is asserted to result in 300 texts, 100 per pool, with no duplicates.

Because the pools differ in size, so do the sampling fractions. The design puts equal n per category ahead of proportional representation, so that category comparisons are not confounded with unequal sample size.

What is and is not reproducible. The classification rule is fully reproducible and is verified against all 300 sampled texts. The *specific membership* of the sample is not recoverable from the seed alone, because it depends on the prior sample’s contents and

on the candidate ordering at the time each replacement was drawn. `sampled_texts.csv` is therefore the authoritative record of which texts were used.

Seven verification checks on the sample (size, balance, absence of reserved examples, absence of duplicates, pool eligibility, agreement with the classification rule, and absence of off-label texts) all pass and are enforced as assertions rather than advisory output in `00_corpus_and_sampling.ipynb`.

One limit of the off-label filter. Step 0 screens two labels per text, the author’s own and the selected reader-majority label, not the individual reader votes. A text can therefore enter the sample while still containing reader votes outside the eleven-emotion set, and 27 such votes survive across 25 texts, so **275 of the 300 sampled texts retain all five in-inventory reader votes**. Those votes are excluded when the original-corpus vectors are built, which is why vector construction tracks n per vector rather than assuming five throughout (Section 4.3.1).

3.4 Ambiguity Types and Author Relevance

Not all ambiguous texts are ambiguous for the same reason. Some are uncertain because author-related context is missing; others describe an underspecified or emotionally complicated situation in which the author’s identity is not material. H3 relies on this distinction, since it predicts that framing effects concentrate in the former.

Texts were accordingly sorted into three groups, used consistently throughout the thesis:

- **Author-relevant ambiguous texts**, where information about the author’s role, perspective, or position within society could plausibly shift the perceived emotion. A description of a risky action may read as pride from the agent who took it and as fear or worry from a spectator.
- **Author-independent ambiguous texts**, where several emotions remain plausible regardless of who wrote the text, typically attributable to a lack of situational detail rather than to an authorship issue.
- **Unambiguous control texts**, drawn from the high-agreement pool, where the writer and the readers agree strongly on one emotion.

The final sample consists of **300 texts, balanced at exactly 100 per category**. This balance is a property of the sampling design and is verified programmatically as part of the quality-control gate described in Section 4.5.

One property of this categorisation constrains what H3 can claim, and it must be stated plainly. Texts were assigned to the author-relevant group partly on the basis of an LLM judge’s own sensitivity to whether author framing would invert the reading: the corpus preparation records, for each text, two candidate persona readings from that judge and whether they differ. Texts were therefore *selected* into that group partly on the property H3 subsequently measures. H3 is consequently a **transfer comparison**: it asks whether a grouping based on the judgments of one LLM predicts framing effects in newly collected human data. That author-relevance is the operative principle is therefore not an independent discovery, and Chapter 5 does not present it as one.

The impact is sharper on the model side than on the human side. Because the category itself was defined by whether an LLM’s judgment flips between personas, the finding that the *model ensemble’s* framing-effect magnitude is ordered author-relevant > author-independent > unambiguous (Section 5.9.1) is partly true by construction: texts were selected into the author-relevant pool precisely because at least one LLM’s reading was sensitive to the persona pair. The ordering holds across all six prompt configurations, and for an ensemble of seven distinct models rather than just the one used as classification judge, which suggests that the pattern is more than an artefact of the selection. Nonetheless, the model-side ordering claim must be read as partially circular rather than as an independent confirmation, just as the human-side H3 result is treated as a transfer comparison rather than a fresh discovery.

3.5 Author-Framing Manipulation

Perceived authorship is the independent variable. The content of every text is held *exactly* constant; only a short author cue attached to it varies, so no difference in emotion judgment can come from a change in the text itself. The cue is appended after the text rather than placed before it (Section 4.1.1 gives the exact format). Which side of the text the cue sits on is not expected to matter for the manipulation’s logic, but the earlier description of it as *preceding* the text was not consistent with how it was actually presented. Nor does the manipulation by itself isolate identity from every other kind of information the cue carries: a persona description can also supply substantive situational

detail beyond who the author is (for instance, a persona contrasting someone who has been struggling all year with someone expecting to pass easily differs in circumstance as well as in identity), so the design measures sensitivity to the provided author/persona framing as a whole, not a pure identity mechanism in isolation.

Four framing conditions were used:

- **Neutral:** no author information is provided.
- **Generic human:** the text is attributed to an unspecified human author.
- **Persona A** and **Persona B:** the text is attributed to a person with a defined social role.

The generic human condition is more than an extra level; the analysis depends on it. Neutral and persona prompts differ in more than persona content, since persona prompts include an author line that is absent from neutral prompts. A neutral-to-persona comparison therefore confounds *who the author is* with *whether an author is named at all*. The generic human condition separates the two, and Section 5.10 uses it for exactly that purpose.

Persona selection was not straightforward. There is no single fixed pair, such as *politician* and *comedian*, that applies equally to every text; a domestic anecdote has no natural connection to either role. Personas were therefore assigned per text to suit its subject matter, with the two personas of a pair chosen to have contrasting interpretive expectations (one related to accountability, one to latitude, for instance), so that a framing effect, if present, has a reasonable chance of emerging. Both personas for every text, along with the reasoning behind the pairing, are recorded explicitly in the dataset.

3.6 Realised Experimental Design

Table 3.1 summarises the design as executed.

Human annotation. Participants were recruited and presented with texts one at a time together with the assigned author framing, selecting a label from the same fixed set of eleven emotions used for the models. Texts and framings were distributed such that each text–framing pair received several independent annotations from different people, while **no individual annotator saw the same text more than once**. This property was specified in the approved proposal, and the statistical analysis of the human data in

Table 3.1: Realised experimental design.

Factor	Levels	Count
Texts	balanced across 3 ambiguity categories	300 (100 per category)
Emotion labels	fixed, alphabetically ordered	11
Framing (models)	neutral, generic human, persona A, persona B	4
Framing (humans)	neutral, persona A, persona B	3
Models	7 (proprietary, commercial and open-weight)	7
Prompt variants	zero-shot, one-shot, few-shot	3
Label formats	numbered list, inline	2
Model judgments	$300 \times 4 \times 3 \times 2 = 7,200$ cells, each an ensemble vector aggregating up to 7 models' raw answers (50,400 raw answers total)	7,200 ensemble vectors
Human judgments	300×3 sides	900 vectors

Section 4.6 is based on it: since each participant was shown only one framing of a given text, the conditions for a text are rated by disjoint sets of individuals, and there is no within-text pairing across conditions to preserve.

Model evaluation. Each model saw each text only once per framing condition and prompt configuration, avoiding repeated exposure. The predicted label for each (text, framing, variant, format, model) combination was recorded. Prompt configurations are always kept separate in any analysis; they are only reported, or averaged, after being calculated independently.

3.7 Deviations from the Approved Proposal

The departures from the proposal are detailed here so that none is left implicit.

Sample size. The proposal expected 300–400 texts. The actual sample is 300, chosen to allow an exact balance of 100 per category.

Number of models. The proposal outlined at least one proprietary, one commercial, and one open-source model. Seven were used, which substantially supports RQ2: a framing effect that shows up across seven independently developed systems is more difficult to attribute to a peculiarity of a single model family.

Number of prompt configurations. The proposal specified at least two prompt variants. Three variants were crossed with two label formats (six configurations), making it possible to evaluate prompt sensitivity as an exploratory question in its own right.

Ambiguity typing and persona mapping. The proposal envisaged manual inspection of a subset of texts, thematic clusters, and persona pairs assigned consistently within each cluster. The realised classification instead employed a fixed consensus threshold and a documented pilot persona-flip decision for the remaining texts; persona pairs were assigned text by text. The category definition incorporates the pilot judgments, so the author-relevant category is partly selected for the property tested by H3 (Section 3.4). Under this per-text design, a cluster-to-persona mapping was not generated. The realised pairings and rationales can be inspected only in the master sampling file; they are not catalogued in this manuscript.

Asymmetry in framing conditions. Models were evaluated under all four framings; humans were annotated under three, omitting the generic human condition. The control analysis in Section 5.10 can therefore only be applied to models, and it is not possible to provide a parallel human decomposition of author-presence versus persona-content. Section 6.3 returns to this limitation of the collected data.

Inter-annotator agreement. The proposal specified Fleiss’ κ [Fleiss 1971]. This statistic does not require respondent identifiers: each item contributes its counts of emotion votes. Since the realised number of votes per (text, framing) cell ranges from three to seven, the conventional equal-rater calculation is restricted to the 754 cells with exactly three votes. The other 146 cells are excluded from this descriptive calculation only, not from the main distributional analysis. Kappa is calculated separately within each ambiguity pool and framing condition across the eleven emotion labels (Section 5.19). It corrects for the group-specific marginal emotion frequencies, so comparing values between groups is a descriptive rather than a causal claim about framing. The equal-three restriction can also select a non-representative subset of persona cells.

Model evaluation metric. The proposal specified macro-F1 and percentage agreement against human majority labels. Section 5.4 provides both as a secondary neutral-framing benchmark against the original corpus’s reader majority, using only the 241 sampled texts with a strict majority among five readers. Each of the six prompt configurations is scored individually and averaged by model; invalid labels are removed, and the count of retained predictions is presented in the reproducible table. The benchmark answers a narrower question than the main analysis: it uses a noisy reader majority as the gold label, filters out 59 texts with no strict majority, and does not score persona responses against the separately collected human votes. Model framing behaviour is instead characterised

distributionally: by agreement with human vectors at baseline (Section 5.6), and by the magnitude and direction of the shift under framing (Sections 5.9, 5.12).

Qualitative case analysis. The proposal outlined an analysis of individual strong-shift and stable-text cases. One traceable example, selected for its observed human shifts and reported in Section 5.14, illustrates how the reading of the same text can change completely under one persona and stay completely unchanged under another. This illustrative example is not a systematically coded thematic analysis of cases and cannot be used to determine why the shift occurred.

Explanations and corpus summaries. All seven analysed models provided a nonempty explanation for every one of their 7,200 responses (coverage is documented in `results/model_explanation_coverage_verified.csv`); one such response is discussed in Section 5.14. Neither an explanation-based thematic analysis nor a model confidence analysis was conducted. Section 5.2 provides author-label frequencies and text lengths for the final 300-text sample. These sample statistics are not a proxy for a separately requested descriptive analysis of the full source corpus.

Participant count. The proposal projected 10–15 participants. The realised study used a larger, less formally demarcated pool recruited across multiple collection rounds (an early round and a later persona/neutral re-collection round, Section 4.1.5), with a target of three annotators per (text, framing) cell. Since respondent identifiers were not stored (Section 4.2), the actual number of distinct participants cannot be reconstructed after the fact, and no such claim is made. This is itself a limitation, revisited in Section 6.3. The within-text permutation null of Section 4.6.2 does not require knowledge of the total number of people involved, but different people rating the two conditions eliminates only the within-text pairing requirement; it is not, in itself, evidence that the two conditions are exchangeable (Section 4.6.2).

4 Implementation

This chapter documents the entire analytical pipeline, from prompt construction to the statistical procedures that underpin each hypothesis test. The aim is reproducibility, so every transformation applied to the data is explained, along with the decision made wherever more than one treatment was defensible.

The pipeline has five stages. Prompts are constructed and models are queried (Section 4.1); human annotations are gathered under the same materials (Section 4.2); every judgment, human and model alike, is converted into an eleven-dimensional probability vector (Section 4.3); metrics of shift are derived from those vectors (Section 4.4); and a quality-control gate (Section 4.5) must be passed before any inferential statistic is calculated (Section 4.6).

4.1 Prompt Construction and Model Evaluation

4.1.1 Stimulus Construction

Each stimulus consists of the unmodified text followed by an optional author cue, in the fixed format "Text: {text}" for the neutral condition and "Text: {text}\nAuthor: {cue}" for the other three, where {cue} is the literal string "a person" for the generic human condition and the assigned persona description for the two persona conditions. All four framing conditions keep the text byte-identical; only the cue, and whether it appears at all, varies.

One property of the source corpus had to be fixed during implementation. crowd-enVent texts were elicited using the frame "*I felt [emotion] when/that/if...*", and in some texts the author's own emotion word was still present in the stimulus. A text like this gives its own answer away. In every affected text, the emotion word was redacted and replaced with an ellipsis, which keeps the sentence structure but removes the answer. Section 4.1.5

documents how the issue was detected and corrected after an initial round of model collection.

4.1.2 Model Selection

Seven large language models were tested, spanning proprietary, commercial and open-weight systems: **OpenAI**, **Claude**, **Gemini**, **Qwen3**, **Gemma3**, **Ministral** and **Llama 3.1**. Proprietary and commercial models were queried through their respective APIs; open-weight models were run locally through Ollama. The shared collection script also supported a Moonshot Kimi K3 endpoint as a fourth proprietary/commercial option, but no Kimi K3 output was generated or retained: the corresponding `KimiK3_label/` `KimiK3_reason` columns are empty across all 7,200 rows of the raw result files, and a dedicated quality-control check confirms Kimi K3 has no normalised-label column and is correctly absent from the seven-model set analysed throughout this thesis.

Recovered model identifiers. The defaults set in the collection scripts (not recorded per-row in the result files themselves, so recovered from the scripts and their run logs rather than from the data) give: `gpt-5-nano` (OpenAI), `claude-haiku-4-5-20251001` (Claude), `gemini-3.6-flash` (Gemini), `qwen3:8b`, `gemma3:4b`, `ministral-3:8b` and `llama3.1:8b` (open-weight, via Ollama). No explicit temperature, top-*p*, or other decoding parameter was set for any model, so each used its provider’s or Ollama’s own default, which is not separately recorded and cannot be recovered. Collection dates can only be reconstructed partially: run logs of the open-weight models date their collection to early September 2026 (specifically 2026-09-03 to 2026-09-04); no equivalent timestamped log was found for the proprietary and commercial models’ collection runs. These identifiers are the defaults actually present in the scripts that produced this data; they are not independently confirmed against a per-response API version field, since none was retained.

The proposal required three systems; seven were used because of RQ2. What one model shows as a framing effect could be an artefact of a single training pipeline, but an effect that replicates across seven independently developed systems is far less plausibly attributed to any single one of them.

4.1.3 Prompt Configurations

Six prompt configurations were used, formed by crossing three prompt variants with two label formats:

- **Variants:** zero-shot, one-shot, few-shot.
- **Label formats:** the eleven emotions presented as a numbered list, or embedded inline in the question.

Each model answered every (text, framing) pair once under each configuration, giving $300 \times 4 \times 3 \times 2 = 7,200$ cells per model-ensemble position. Configurations are **never pooled at the level of raw responses**. Where an aggregate across configurations is given, each configuration’s results are calculated independently first and then averaged, so that no analysis treats six correlated observations of one text as six independent observations.

4.1.4 Response Normalisation and Off-List Answers

Models were instructed to return exactly one emotion from the fixed set of eleven, and not every response did. Two kinds of non-compliance were handled explicitly.

Formatting variation, answers that differ only in case, whitespace, or surrounding punctuation, were normalised to the canonical label.

Off-list answers, where the respondent named an emotion outside the eleven, such as *anxiety*, *disappointment*, or *frustration*, were **excluded** rather than mapped onto the nearest in-set label. Mapping would mean deciding which of the eleven a given synonym corresponds to, and that decision would itself be an interpretive judgment inserted into the data by the analyst. Exclusion is the conservative choice: it shrinks the effective ensemble for the affected cell instead of inventing a vote.

Off-list rates differ substantially by model and are reported in Section 5.3. The present analysis does not record any synonym-mapping sensitivity result; whether the exclude-only policy affects a specific conclusion remains untested here.

4.1.5 A Data-Quality Correction Made During Analysis

Corrections made partway through the study are documented, rather than presenting the corrected data as though it had always existed. The emotion-word leak described in Section 4.1.1 was caught and fixed on two separate occasions with different scopes. Both are described below, not only the more visible second one.

First round. An early audit of the stimulus texts identified 401 candidate texts still carrying the author’s emotion word, of which 102 fell inside the 300-text sample (78 corrected in place, 24 replaced outright because a clean substitute was not otherwise available). The model side was fully re-run against the corrected corpus. Since the human side had already been annotated for 78 of those 102 texts before the fix, those human ratings answered a stimulus that no longer existed; they were discarded and re-collected, together with the human annotations for the 24 replacement texts, in a dedicated collection round. The same round also collected the study’s first human ratings for the neutral condition, which up to that point had been supplied by reusing the original corpus’s five-reader labels, a different, disjoint rater pool from the persona conditions. All of this, 900 human vectors across three framings, each now from data collected specifically for the post-correction 300-text sample, was in place before the second round began.

Second round. After an initial round of *model* collection against that corrected corpus, a further 49 of the 300 texts (16.3%) were found to still retain the author’s emotion word in the model stimulus, and the same texts additionally carried an outdated persona pair from an earlier design revision. The corrected wording existed in the master sampling file but had never been propagated into the model prompt files for that batch. This is a distinct error from the first round rather than a repetition of it: it affected the *model* prompt files specifically, was not identified until the human side described above was already complete, and does not claim to share a common cause with the first round beyond both having occurred within the same elicitation frame.

The error was corrected through **full re-collection**, not through exclusion. All 1,176 affected rows (49 texts $\times$ 24 conditions) of the stimulus and both personas were reconstructed from the corrected master file, previous answers for those rows were discarded since they responded to a prompt that no longer existed, and all seven models were re-queried.

The change in results was material, and that is the reason for reporting it. Model persona sensitivity rose from 8.7% to 11.3% on unambiguous texts and from 12.2% to 14.8% on author-independent texts, while author-relevant texts, of which only nine were affected, shifted only marginally. The size of this change follows the number of affected texts per category, consistent with the correction having removed a suppressing artefact. This round, however, altered two things at once for the affected 49 texts, both the leaked emotion word and the persona pair itself, so the rise cannot be credited to removing the leak alone. Some or all of it may instead reflect the updated persona pairs being more effective manipulations than the outdated ones they replaced. No hypothesis verdict changed qualitatively.

4.2 Human Annotation Study

Human annotators saw the same texts, the same eleven-label set and the same author cues as the models, so both sides were responding to materially identical stimuli.

Each (text, framing) pair was annotated several times by independent annotators, with the aim of three annotators per cell; some cells had more. As the approved proposal specified, **no annotator read the same text more than once**. A participant who rated a text under one framing never saw that text under a different framing.

The statistical treatment in Section 4.6 rests on this property. Since the conditions of a given text were rated by disjoint sets of individuals, there is no individual-level pairing across conditions to preserve. The comparison is therefore **between-subjects at the level of framing condition**. Disjoint raters are a documented property of the collection procedure; exchangeability under a null of no framing effect additionally requires that the groups assigned to each condition would, absent a framing effect, have produced the same distribution of judgments, which disjoint recruitment alone does not guarantee if allocation or the participant pool differed systematically across rounds (Section 4.6.2).

The neutral condition was collected in a separate, later round from the persona conditions. Only vote values were stored in the analysis pipeline; respondent identifiers were not retained. The consequences of this for inference are stated in Section 4.6.

A traceability limit in the neutral collection. The neutral condition’s forms were initially designed and distributed for one rater per text, and the collection records available alongside this thesis’s data reflect that design. The dataset used throughout this

thesis instead carries three raters per neutral cell, matching the persona conditions. The source of the additional two raters per text for the neutral condition could not be reconstructed from the retained collection records at the time of writing. It is disclosed as a limit on this thesis’s own reproducibility, not as an allegation that the three-rater neutral vote counts are invalid: these are exactly the numbers every downstream calculation in this thesis, including all six H1 tests, is computed from, and nothing in the analysis pipeline is inconsistent with them. But a reader who wants to trace every human vote back to a raw, timestamped response should know that this particular step in the collection history is not fully documented in the materials retained for this thesis.

4.3 Emotion Probability Vector Construction

The representation described here is the core of the thesis.

4.3.1 Definition

Let $E = \langle e_1, \dots, e_{11} \rangle$ be the fixed, alphabetically ordered emotion set. For any group of judgments about a single (text, condition), whether those judgments come from human raters or from models, let c_i be the number of judgments naming emotion e_i , and let $n = \sum_i c_i$ be the number of usable judgments. The emotion probability vector is

$$\mathbf{v} = \left\langle \frac{c_1}{n}, \frac{c_2}{n}, \dots, \frac{c_{11}}{n} \right\rangle, \quad \sum_{i=1}^{11} v_i = 1. \quad (4.3.1)$$

Every vector in this thesis is constructed this way. A human vector for a (text, framing) pair aggregates that cell’s raters; a model ensemble vector for a (text, framing, variant, format) cell aggregates the seven models’ single answers. Off-list and missing answers are excluded from c_i and reduce n accordingly; they are never silently counted as zeros, which would distort the denominator.

4.3.2 Why Distributions Rather Than Labels

The reason for this representation is empirical. In the original five-reader corpus, only 80.3% of texts have a unique emotion commanding a strict majority. Of the remainder, 6.3% have a unique top choice that never reaches 50% (a plurality, not a majority) and

13.3% have **no unique top choice at all**. For roughly one text in five, a pipeline that reports a single label must either retain the most frequent label while concealing how divided readers actually were, or break a tie arbitrarily.

A vector sidesteps the choice. It can also express something a label cannot: the difference between a judgment that barely moved and one that was substantially reinterpreted but happened to keep the same most-popular answer. Section 5.6 measures that difference.

4.4 Metrics

The vectors give rise to two families of metric: one for the *magnitude* of a shift and one for its *direction*. They answer different questions and are never conflated.

4.4.1 Magnitude: Total Variation Distance

For two vectors $\mathbf{p}$ and $\mathbf{q}$ over the same emotion set, total variation distance is

$$\text{TV}(\mathbf{p}, \mathbf{q}) = \frac{1}{2} \sum_{i=1}^{11} |p_i - q_i| \in [0, 1]. \quad (4.4.1)$$

TV is the share of probability mass that must be moved to turn one distribution into the other. It is zero when the two are identical and one when their supports are disjoint. Shift magnitudes are reported as $\text{TV} \times 100$ and referred to as shift percentages.

TV measures magnitude only. It expresses how much moved, but not which emotions moved or in which direction; the directional metrics below are therefore reported alongside it as a matter of course, not as an optional extra.

4.4.2 Direction: Delta Vectors and Cosine Similarity

The signed change induced by framing is the delta vector

$$\boldsymbol{\delta} = \mathbf{v}_{\text{persona}} - \mathbf{v}_{\text{neutral}}, \quad (4.4.2)$$

whose components sum to zero and record which emotions gained mass and which lost it. To ask whether a model's shift resembles the human shift *for the same text*, the two delta vectors are compared by cosine similarity:

$$\cos(\delta_H, \delta_L) = \frac{\delta_H \cdot \delta_L}{\|\delta_H\| \|\delta_L\|}. \quad (4.4.3)$$

Cosine is undefined when either delta is the zero vector, which occurs whenever one side did not move at all. These cases are **counted and reported** rather than silently dropped, because their frequency is itself a finding (Section 5.16).

4.4.3 Dominant Sets and Their Five Outcomes

The dominant set of a vector is the set of emotions attaining its maximum probability. It is a *set* rather than a single label because ties are common at small sample sizes and resolving them arbitrarily is exactly the practice this thesis is examining.

Comparing dominant sets before and after framing yields five descriptive outcomes:

- the sets are **identical**;
- a single emotion **splits into a tie**;
- a tie **resolves** to a single emotion;
- two ties **partially overlap**;
- the sets are **disjoint**.

These describe relationships between complete top sets. A selected-label analysis instead chooses one winner from each set, so its changed/unchanged verdict does not coincide with any one of these five outcomes; a tie can change the chosen winner while the sets still overlap. The set outcomes are given in Section 5.17 and the selected-label comparison in Section 5.6.

4.4.4 Dispersion Measures

Three further quantities summarise the shape of a vector: Shannon entropy (and its normalised form), maximum probability as a measure of consensus strength, and support size, the count of emotions receiving any mass. These address a question distinct from

magnitude, namely whether framing *concentrates* or *disperses* interpretation, and are examined in Section 5.15.

4.5 Quality-Control Gate

Every inferential statistic in this thesis was computed only after its inputs passed quality control. Since the notebook rebuild described in Section 4.6.6, this rule is enforced at two levels rather than through a single external report. First, the data-loading routine that every notebook calls halts immediately, by assertion, if a structural invariant fails: there is no code path by which an analysis can run against malformed inputs. Second, an independent verification script re-derives 30 scientific and execution checks from the raw inputs and confirms that all three notebooks executed top to bottom with zero errors and consecutive execution counts; all 30 pass.

The checks fall into four groups:

- **Completeness:** every text present in every derived table; expected row counts exactly met (900 human vectors yielding 600 comparisons across two persona sides, 3,600 model comparisons across two persona sides and six prompt configurations, exactly six configuration rows per text and persona side).
- **Mathematical validity:** every probability vector sums to one; every delta vector sums to zero; total variation equals the sum of positive signed change; the retained vote count for every model comparison is equal on both sides of the comparison.
- **Design integrity:** exactly 100 texts per category; at least three model identities common to both sides of every model comparison; no duplicate (text, condition) rows; every one of the 300 sampled texts reproduces the classification rule of Section 3.3.1 exactly against the 985-text eligible candidate pool.
- **Source consistency:** every category value maps to a known category, and no text is classified inconsistently across its own rows.

The final group was written to catch a particular bug. The human annotation file disagrees with itself on nine texts: one category is recorded in the neutral rows and another in the persona rows, a relic of a reclassification between rounds of collection. Because naive de-duplication does not remove such a conflict but silently resolves it by whichever row happens to come first, those nine texts had crept into the human-side

analyses under the wrong category, producing an unbalanced 91/100/109 grouping in place of the designed 100/100/100. The fix is structural: instead of reading the per-row category column of the human annotation file, the master sampling file is read once, at data-loading time, and the category of every text is looked up from it. The nine texts are checked explicitly to confirm that the defect is real and has been correctly resolved, not to certify a list that could otherwise remain stale.

4.6 Statistical Procedures

This section sets out the inferential procedures and, just as important, the unit of analysis for each test.

4.6.1 Unit of Analysis

The risk that keeps recurring in this dataset is pseudoreplication. A single text provides 2 human comparisons and 12 model comparisons (2 persona sides $\times$ 6 configurations). Treating those as independent observations inflates n by as much as twelvefold, and correspondingly inflates significance.

Accordingly, unless stated otherwise, **the unit of analysis is the text**. Where a statistic is computed on finer-grained rows, the confidence interval is instead obtained by a bootstrap that resamples texts rather than rows, so that the correlation among a text’s own observations is preserved in the resampling.

4.6.2 Permutation Tests

Two different permutation nulls are used [Good 2005], and the difference between them is one of substance, not merely of technicality.

For the human data (H1), the votes are pooled within a text across the two conditions under comparison, reshuffled, and re-split at the observed group sizes. The different conditions of a text were rated by different people, so there is no within-text participant pairing requirement. The permutation also presumes that, given no framing effect, the groups assigned to that text’s conditions would have the same distribution of emotion judgments; disjoint participants alone do not imply exchangeability if recruitment or allocation differed systematically.

For the model data (H2), the same pooling would be invalid. The same seven models answer both conditions, so model identity is tracked on both sides, and the models are not exchangeable with each other; Section 5.9.1 measures directly how differently they behave. The correct null holds model identity fixed and asks whether each model’s own two answers could just as well have been obtained the other way round. The test is therefore a **paired sign-flip**: for each (cell, model) pair independently, the two labels are swapped with probability one half.

Both tests use 10,000 iterations with a fixed random seed.

4.6.3 Effect Sizes and Confidence Intervals

Effect sizes are reported alongside every p -value. Category contrasts (H3) are reported as the raw difference in mean shift between groups, with a text-level bootstrap confidence interval, rather than as a standardised ordinal effect size; matched-pairs rank-biserial correlation is used for paired comparisons (H4 magnitude), and Spearman’s ρ for monotonic association (H4 covariation). Confidence intervals are percentile bootstrap intervals with text-level clustering.

Where an association is summarised by a correlation, inference is drawn from the **clustered bootstrap itself**, as opposed to an analytic p -value computed over all rows. Over correlated rows, an analytic p is not merely imprecise but misleading. Duplicating every row would leave ρ unchanged and shrink p , the very behaviour clustering is meant to prevent.

4.6.4 Multiple Comparisons

Two correction regimes are used, and which applies is indicated at each test. Confirmatory hypothesis families are controlled by **Holm** correction [Holm 1979], where the family-wise error rate should be controlled. Exploratory families, such as the 66 permutation tests (Section 5.13), where the aim is to limit the proportion of false positives among discoveries rather than exclude them entirely, are controlled by **Benjamini–Hochberg** false discovery rate control [Benjamini and Hochberg 1995].

4.6.5 Referencing Against a Sampling Null

One last procedure is needed because of an easily overlooked property of small-sample TV.

TV between two small vote samples is not zero when there is no effect.

Two independent three-vote draws from the same underlying distribution will often disagree, and they disagree more when the underlying distribution is dispersed. Since the ambiguous categories are by construction the dispersed ones, a raw comparison of TV across categories conflates baseline dispersion with any framing effect.

To separate the two, each text is given its own reference point. That text's votes are pooled across the two conditions, reshuffled, and re-split at the observed sizes; the average TV over many such draws is the TV expected for that specific text if framing did nothing. The observed value minus this expectation is referred to as the text's **excess TV**.

This is applied in Section 5.11.2 as a sensitivity analysis. It is reported next to the raw contrasts and never replaces them: it was specified after the raw result was known, and making it the primary analysis would amount to selecting an analysis by its outcome.

4.6.6 Reproducibility

All permutation tests and bootstrap operations are deterministic and were confirmed to reproduce exactly across separate process invocations with a fixed seed. A single non-determinism bug was discovered and fixed during development: an unsorted set intersection produced an iteration order dependent on per-process hash randomisation, causing seeded results to be irreproducible across runs. All collections of string keys feeding a seeded loop are now explicitly sorted before use.

The analysis notebooks were later rewritten so that the calculations behind the four main hypotheses are stored directly within the notebook rather than being imported from a separate module; the core pipeline can now be audited without referring to other files. All 44 result tables generated by the rewritten notebooks were compared against the results generated prior to the rewrite and were binary equivalent; the rewrite reformatted, but did not redefine, the calculations. A small number of additional results added later (the inter-annotator agreement figures, the conventional majority-label benchmark, the entropy hypothesis tests of Section 5.15, and the transition-matrix figure of Section 5.17)

are not inlined into the notebooks but are generated by independent scripts alongside them.

5 Results

The chapter is ordered by the question that motivated each analysis. It starts by treating each text and condition as the choice of a single emotion and asks what this approach tells us about H1–H4. It then shows the changes this single label misses. Finally, it reruns the hypothesis tests on the complete eleven-emotion distribution and examines which emotions gained or lost support. The statistical tests ask whether a salient pattern can be distinguished from the variation one would expect simply because only a few people rate each condition; they are not a substitute for the substantive question.

The executed analysis notebooks provide the original analysis numbers; the subsequent agreement, corpus-description, explanation-coverage and text-level correlation checks are reproducible via scripts and CSVs in the accompanying package. Every test of significance is reported with an effect size, and non-significant results are reported rather than omitted.

5.1 Disagreement Is Present Before Any Manipulation

This thesis is based on the premise that judgments of emotion often have no single right answer [Pavlick and Kwiatkowski 2019; Uma et al. 2021]. Checking that this holds for the source corpus is the first step, before any experimental manipulation is applied.

Each vector was categorised into exactly one of four mutually exclusive vote patterns: *unanimous* (all judgments on one emotion), *majority* (one emotion has a unique, strict majority of more than 50%), *plurality without majority* (one emotion has the unique top share but no more than 50%), and *tied* (two or more emotions share the top share).

Interpretation. Among the sampled texts, 80.3% have a unique emotion with a strict majority (71 unanimous and 170 non-unanimous majority texts, i.e. 241 of 300). This sample is stratified two-thirds towards ambiguous categories by design (Section 3.3.2), so this rate does not characterise the large source crowd-enVent corpus, only the sampled

Table 5.1: Vote-pattern distribution across the three judgment sources.

Pattern	Original corpus (up to 5 retained votes, 300 texts)	New human (900 vectors)	Model ensemble (7 models, 7,200 vectors)
Unanimous	23.7%	18.3%	36.8%
Majority (non-unanimous)	56.7%	49.0%	53.3%
Plurality, no majority	6.3%	4.8%	5.8%
Tied	13.3%	27.9%	4.1%

texts studied here. The remaining **19.7%** lack a strict majority: of these, 6.3% still have a unique plurality label, so a pipeline reporting a single label would apply a plurality rule that overstates agreement, while the remaining 13.3% have no unique top choice at all and would require a tie to be broken arbitrarily.

Two more points stand out. The new three-rater human data are tied more often (27.9%): this follows arithmetically from small n and is not a sign that the new annotators disagreed more. The model ensemble is least tied (4.1%) and unanimous most often (36.8%): the seven models are much more likely to converge than human readers, a difference that recurs throughout this chapter.

5.2 Sample Composition

In the final sample of 300 texts, the original author’s emotion labels occur as follows: anger 32, disgust 35, fear 27, guilt 14, joy 25, pride 21, relief 25, sadness 29, shame 19, surprise 33 and trust 40 (`results/sample_author_emotion_frequencies_verified.csv`). Text length has a median of 13 whitespace-separated words (interquartile range 8–25; minimum 2, maximum 203). These are *sample* summaries, filtered by category balancing and eligibility vetting, not frequencies for the original full crowd-enVent corpus.

5.3 Model Response Validity

Of 50,400 individual model judgments ($7 \text{ models} \times 7,200 \text{ cells}$), **609 (1.21%)** were excluded as off-list or missing. Exclusion rates vary by an order of magnitude across models. These numbers are calculated directly from `experiment_results_all.csv` using the same label-resolution logic the analysis pipeline applies (normalisation, followed

by removal of any response outside the eleven-emotion set). They are a property of the input data, not a result of analysis, and are therefore reported here rather than in an analysis results notebook.

Table 5.2: Excluded (off-list or missing) answers by model, out of 7,200 judgments each.

Model	Excluded	Rate
OpenAI	0	0.00%
Gemini	0	0.00%
Ministral	1	0.01%
Qwen3	8	0.11%
Claude	80	1.11%
Gemma3	93	1.29%
Llama 3.1	427	5.93%

Interpretation. Llama 3.1 accounts for 70% of all exclusions, most often by answering with a near-synonym outside the eleven-label set: *anxiety*, *disappointment*, *frustration*, *embarrassment*. This reflects a real difference in how the models follow instructions, not a data defect. Because off-list answers are excluded rather than mapped to the closest label, they shrink the effective ensemble for the affected cell, so the analyst introduces no interpretive judgment; the cost is a slightly smaller n for those cells.

5.4 Conventional Majority-Label Benchmark

So that the classification metrics named in the proposal are available next to the distributional analysis, each model’s neutral response is compared with the original corpus’s stored reader-majority label. Of the 300 sampled texts, a strict majority (at least three of five readers) exists for 241; the other 59 are excluded so that a tie or non-majority plurality is not silently treated as a gold standard. Percentage agreement and eleven-class macro-F1 are calculated on the valid model answers within each prompt configuration and then averaged equally over the six configurations. Each value in Table 5.3 is therefore descriptive rather than a test comparing models; per-configuration scores and retained counts (217–241 of 241) appear in `results/neutral_majority_label_benchmark_verified.csv`.

These numbers show agreement with one deliberately narrowed, relatively consensual neutral reference set. They do not quantify similarity of full emotion distributions, generalise to the 59 excluded texts, or test whether a model’s framing response matches the human shift.

Table 5.3: Neutral model performance against the original readers’ strict majority, averaged over six prompt configurations.

Model	Agreement	Macro-F1
OpenAI	77.0%	0.742
Claude	73.8%	0.715
Gemini	78.8%	0.764
Qwen3	70.2%	0.686
Gemma3	63.2%	0.585
Ministral	62.3%	0.611
Llama 3.1	68.8%	0.650

5.5 First Pass: H1–H4 with One Selected Emotion

The old approach chooses one top emotion for each condition (using a deterministic tie-break), then records whether that selected emotion changes when a persona is supplied. The question it answers is a plain one: *did the reported label change?* It cannot detect smaller redistributions among the eleven possible emotions.

Table 5.4: First-pass conclusions using selected emotion labels only. The full calculations are in the accompanying old-method CSVs.

Question	Selected-label result
H1, people	None of the six category-by-persona tests is significant after Holm correction. Some labels change, but the evidence is insufficient to detect a human framing effect by this method (<code>old_H1_H2.csv</code>).
H2, models	Both overall persona contrasts are significant ($p_{\text{Holm}} = 0.0002$ each). A model-ensemble effect is detectable even with one label (<code>old_H1_H2.csv</code>).
H3, category	The directly predicted author-relevant minus author-independent difference is 4.50 percentage points and not significant ($p_{\text{Holm}} = 0.472$; <code>old_H3_raw.csv</code>).
H4, alignment	Human label-change frequency exceeds the model-ensemble frequency by 34.44 percentage points; 102 of 438 pairs where both change have the same exact selected-label transition (<code>old_H4_frequency.csv</code> ; <code>old_H4_change_agreement.csv</code>). Direction can be described only as agreement between discrete transitions.

The first pass is informative, and even its negative findings are useful. What remains open is whether it confounds a stable winner with a stable interpretation. The following section measures that loss directly on the same texts and responses.

5.6 What the Vector Representation Preserves

The entire representational choice rests on a claim that can be verified. Each comparison was categorised by what a dominant-label pipeline would have recorded, as well as by what the distribution was actually doing.

Table 5.5: What a selected-label analysis would record. These three outcomes are exhaustive and mutually exclusive; only the middle row is invisible to a label-based pipeline, which records it as unchanged despite the distribution having moved.

Outcome	Human	Model ensemble
No change at all ($TV = 0$)	8.5% (51)	40.2% (1,446)
Moved with the same selected label (proportion-only or support change)	<i>invisible to label analysis</i> 38.7% (232)	41.4% (1,492)
Winner changed (recorded as changed)	52.8% (317)	18.4% (662)

Interpretation. A selected-label pipeline would record 52.8% of human comparisons and 18.4% of model comparisons as changed (`results/old_H4_frequency.csv`). It would additionally file **38.7% of human** and **41.4% of model** comparisons (232 and 1,492 comparisons respectively) as *unchanged*, despite the distribution demonstrably having moved.

For the model ensemble in particular, the number of comparisons invisible to a label-based analysis surpasses the number it detects by more than a factor of two (1,492 versus 662). Here the claim that aggregation eliminates information is quantified across all 4,200 valid comparisons, not argued from examples. A slightly more rigorous, tie-aware version of the same question, asking whether the *whole* set of tied top emotions changed rather than only the alphabetically selected one, yields smaller concealed-movement figures: 25.8% human and 38.7% model (`results/information_lost_by_labels.csv`), since it credits the label method with tracking ties. Section 5.17 reports a related but distinct breakdown, the full set-relationship outcomes between the two sides’ top-emotion sets (identical, disjoint, tie formed or resolved), rather than this same concealed-movement percentage.

Human–model agreement at baseline tells the same story. Under neutral framing, human and model dominant labels agree on 58.0% of comparisons overall, ranging from 74.2% on unambiguous texts down to 49.8%–50.0% on the two ambiguous categories. Within the matching and non-matching buckets, full-vector cosine similarity is distributed widely: for comparisons with matching labels, the mean cosine is 0.90 across the range $[0.41, 1.00]$, and for comparisons with non-matching labels, the mean cosine

is 0.41 across the range $[0.00, 0.98]$. A binary match or mismatch thereby condenses a wide, continuous band of distributional similarity into two buckets, and even within the “agree” bucket the underlying distributions are anything but the same.

Second pass: the same hypotheses with complete emotion distributions. In the following four sections, a shift means that the combination of emotions changed, even though the most typical label may have remained unchanged. A distance of 20% means that one fifth of the total emotion probability would have to be moved to make the two distributions match. Since every human distribution is characterised by only a few votes, some movement naturally occurs by chance even when there is no framing effect. The statistical tests compare observed movement with that baseline; the p -values do not indicate how large or important an effect is.

5.7 Two Independent Neutral Measurements Disagree Substantially

A framing effect is not meaningful until we know how much two measurements of the *same* condition differ. The original corpus’s five-reader neutral round and this study’s own three-rater neutral round are two independent measurements of the same texts under no manipulation.

Table 5.6: Mean TV distance between the original corpus neutral round and this study’s neutral round.

Category	Mean TV
Unambiguous	33.1%
Author-Independent Ambiguous	47.5%
Author-Relevant Ambiguous	53.2%

Interpretation. Two “no persona, no manipulation” measurements of the same texts vary by 33.1% to 53.2% of probability mass, and they vary *more* in the categories that are designed to be more ambiguous. Every shift magnitude that follows has to be read against this noise floor.

The methodological consequence, applied throughout, is that a raw shift percentage cannot be read on its own. A human shift of 45% on unambiguous texts sits barely above a 33.1% baseline disagreement, and it is the permutation control, not the raw

magnitude, that determines whether a framing effect is present. The point returns, more sharply, in Section 5.11.2.

A second, non-overlapping baseline comparison is available for the neutral round specifically: human and model-ensemble dominant labels under neutral framing agree in 74.2% of unambiguous comparisons but only 49.8% and 50.0% of author-independent and author-relevant comparisons respectively, and the corresponding mean TV distances (32.1%, 48.0% and 49.6%) track the same ordering. Humans and models therefore start from noticeably different readings of the ambiguous categories even before any persona is introduced.

5.8 H1: Author Framing Shifts Human Judgments

Hypothesis. Human emotion judgments will differ significantly across author-framing conditions for ambiguous texts, but not (or substantially less) for unambiguous control texts.

5.8.1 Magnitude by Category

Mean text-level shift magnitude rises across the three categories as predicted: Unambiguous **45.1%** (median 41.7%), Author-Independent **54.1%** (51.7%), Author-Relevant **60.3%** (64.6%).

As Figure 5.1 shows, the categories overlap heavily despite this ordering. Interquartile ranges are broad, and every category has individual texts spanning the entire range. The figure does not show that every author-relevant text shifts more than every unambiguous one; it shows a difference in central tendency set against considerable within-category variation.

5.8.2 Permutation Test

Given the noise floor established in Section 5.7, raw magnitudes cannot be interpreted directly, so each category and persona side was tested against a within-text permutation null (10,000 iterations, Holm-corrected across the six tests).

Interpretation. Three results are presented here, in order of evidential weight.

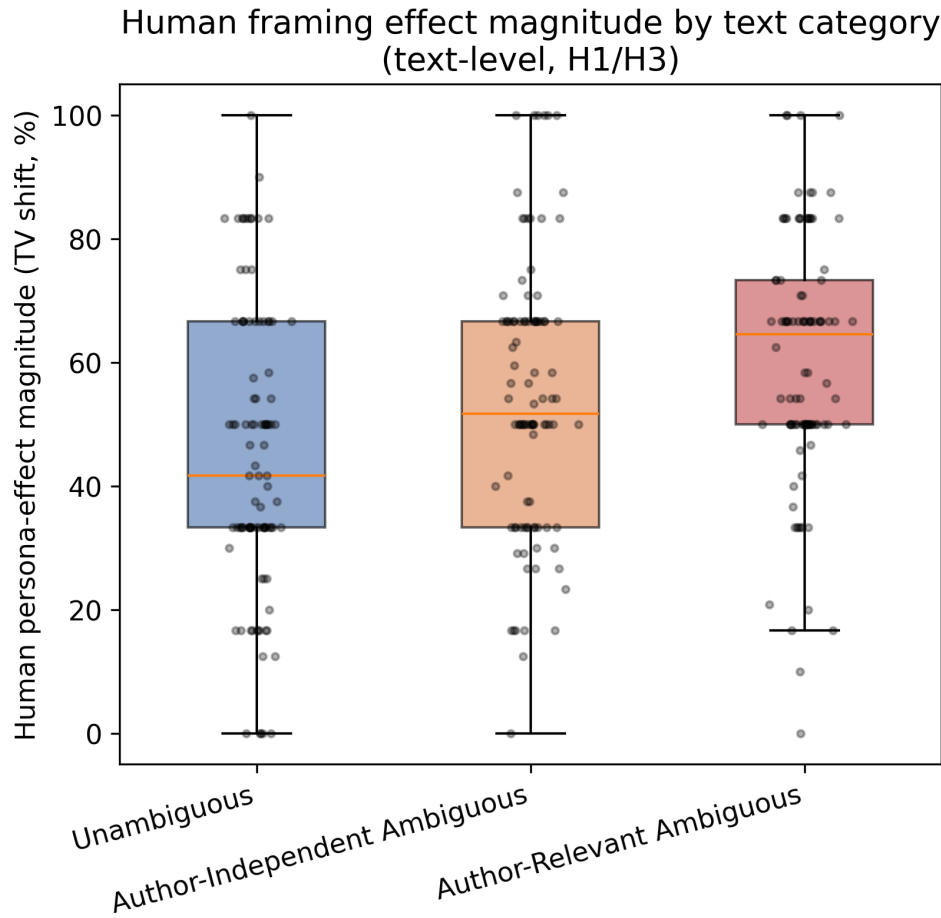


Figure 5.1: Human persona-effect magnitude by text category (text-level, one observation per text).

On author-relevant ambiguous texts the framing effect is significant on **both** persona sides. This is what H1 predicted, and it holds.

On unambiguous control texts there is **no statistically significant** effect on either side. That satisfies H1's control prediction, but non-rejection does not imply that the true effect is zero, or that the manipulation worked only through author content.

For author-independent ambiguous texts, side a crosses the significance threshold but side b does not. The observed shifts are 56.3% and 51.9%, against conditional-null means of 50.9% and 48.9% respectively. *Significant on one side and non-significant on the other is not a test of a difference between sides.* An exploratory direct comparison pairs each text's two raw TV shifts: the mean A-minus-B difference is 4.44 percentage points,

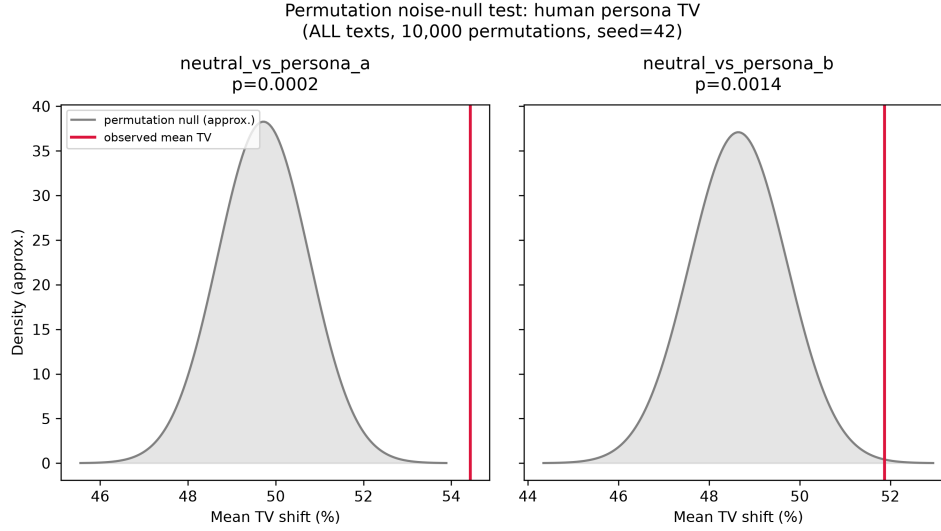


Figure 5.2: Permutation null distributions for the human persona effect, pooled across all texts, with the observed mean marked.

Table 5.7: H1 permutation test results. Holm-corrected across six tests (3 categories $\times$ 2 persona sides).

Category	Persona side a		Persona side b	
	Observed TV	p_{Holm}	Observed TV	p_{Holm}
Author-Relevant	59.9%	0.0066	60.7%	0.0112
Author-Independent	56.3%	0.0110	51.9%	0.1980
Unambiguous	47.1%	0.1980	43.0%	0.2145

with a text-bootstrap 95% interval $[-2.08, 11.06]$ points (paired sign-flip $p = 0.186$; see `results/author_independent_persona_side_contrast_verified.csv`). The evidence on whether *each* framing effect is detectable is therefore mixed, and does not lead to a conclusion that one persona effect is stronger than the other. This exploratory comparison is between raw TV, not between the two separately estimated excess-over-null values.

A cross-check makes the case for the distributional representation empirically rather than by argument. Repeating these same six comparisons on the *selected label* instead of the vote vector (pooling each text’s votes, permuting, and asking only whether the winning emotion differs, with the identical Holm correction) yields **0 of 6** rejections under this thesis’s tie-breaking convention (a vote-share tie goes to whichever emotion is alphabetically first). Under this conventional one-label treatment of the human data, no framing effect is detectable at all, in any category, on either persona side. Although

the respective label-change rates are nonzero (Section 5.6), no corresponding significance test passes once each distribution is reduced to a single winner.

The exact count depends on the tie-breaking rule. Ties are common at these sample sizes, and there is no principled reason to prefer one tie-break order over another beyond the fact that alphabetical order was the one chosen. Repeating the same selected-label test with ties broken in reverse-alphabetical order flips the author-relevant, persona-A cell from non-significant to significant (change rate 73.0%, $p_{\text{Holm}} = 0.0012$), making the overall count **1 of 6** rather than 0 of 6 (`results/tie_break_h1_significance_verified.csv`). This general methodological point, that the selected-label method can at best find only a fraction of what the distributional treatment finds, holds under either convention; the specific claim that it finds *none* of the six does not.

Verdict: H1 is partially supported: supported on author-relevant texts, with no detected effect in the unambiguous controls, and mixed for author-independent texts. This support appears only under the distributional treatment and is absent under the selected-label one; this is one of the chapter’s findings in its own right, taken up again in Section 5.6.

5.9 H2: Author Framing Shifts Model Classifications

Hypothesis. LLM emotion classifications will shift across author-framing conditions, reflecting sensitivity to minimal author cues.

5.9.1 Magnitude by Category and Configuration

The mean ensemble shifts are **11.3%** for unambiguous, **14.8%** for author-independent and **28.8%** for author-relevant texts. The ordering holds for all six prompt configurations (Figure 5.4); all seven models show a positive observed label-change rate. The model ensemble itself is tested with the paired test below, not each model individually.

Individual model label-change rates for author-relevant texts range from Ministral at 39.3% down to Gemma3 at 26.2%. No pairwise significance test was run between models, so these are described as highest and lowest *observed* means, not as best and worst.

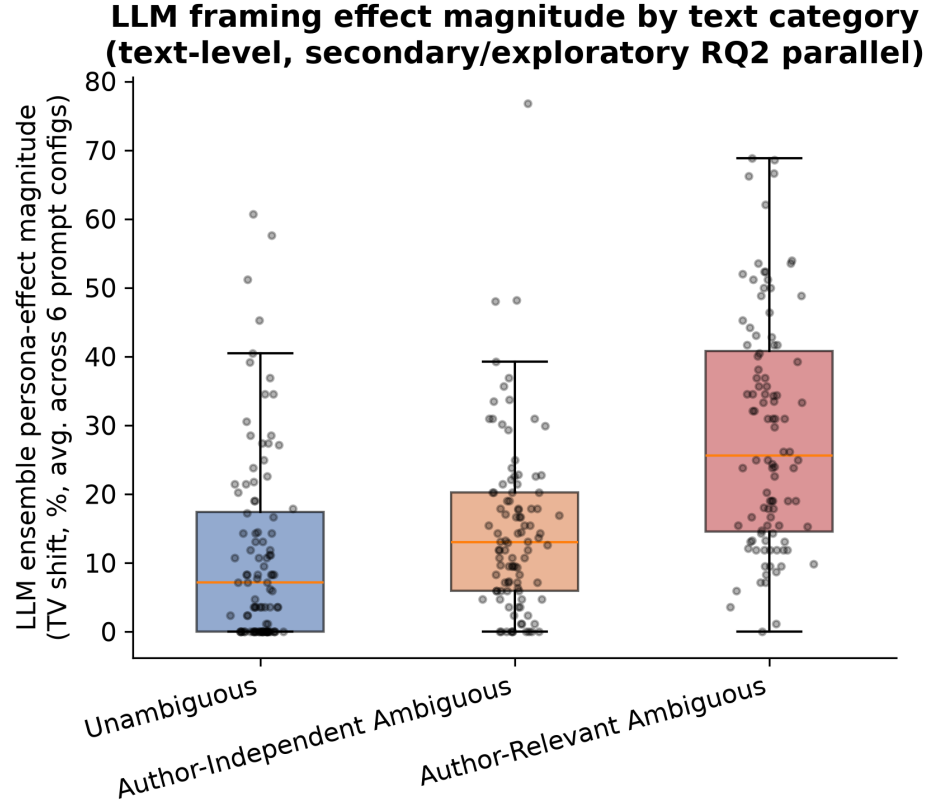


Figure 5.3: Model ensemble persona-effect magnitude by text category (text-level, averaged across the six prompt configurations).

The shape of Figure 5.3 is strongly skewed to the right. The median ensemble shift is substantially lower than the mean, because a large fraction of model comparisons do not shift at all (Section 5.16).

5.9.2 Paired Permutation Test

Because the same seven models answer both conditions, the null must hold model identity fixed (Section 4.6.2). The paired sign-flip test gives:

Table 5.8: H2 paired sign-flip permutation test, Holm-corrected across two persona sides.

Comparison	Observed mean TV	95% CI (text-clustered)	p_{Holm}
Neutral vs persona a	16.80%	[14.74, 18.97]	0.0002
Neutral vs persona b	19.78%	[17.50, 22.08]	0.0002

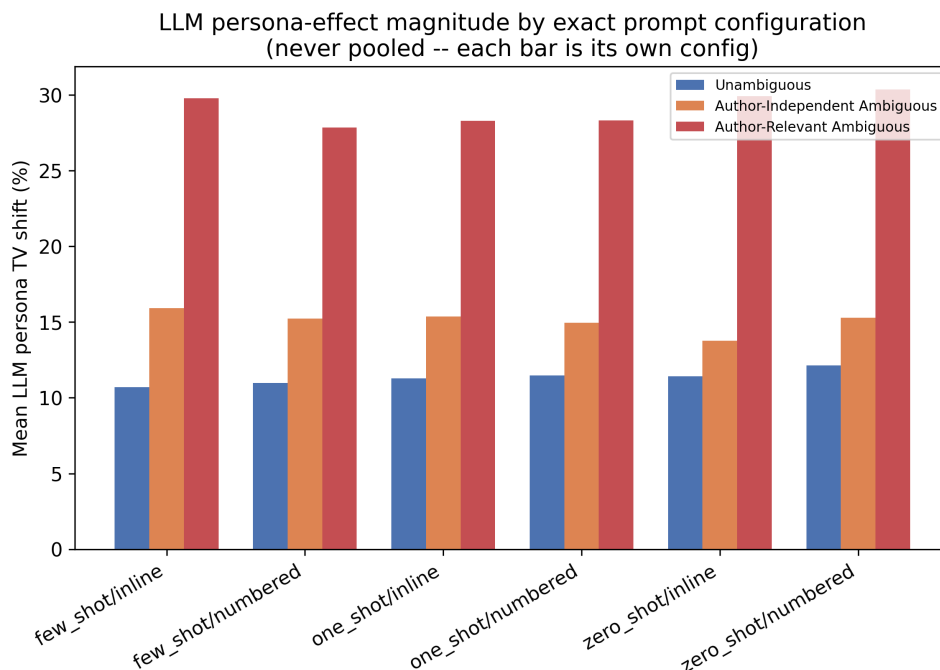


Figure 5.4: Model persona effect by exact prompt configuration. Each bar is a separately computed, never-pooled comparison.

5.9.3 What This Test Does and Does Not Establish

The statistic is TV between ensemble vectors, so the test detects **coordinated net re-distribution** across models. It does not measure whether any individual model changed its answer, and that difference has real consequences.

Under a paired sign flip, concordant model pairs cancel and only discordant pairs contribute. When a cell contains exactly one discordant model, its TV equals $1/n$ under *every* possible swap: the null is a point mass, and the test cannot reject, regardless of how real the change is. In these data, 37.9% of cells have no discordant model and a further 23.3% have exactly one, so **61.2% of cells contribute no variance to the null**. The remaining 38.8% drive the test.

Because of this, the complementary quantity is reported as well: of 24,736 valid model pairs, **5,263 (21.3%) changed their label** between conditions. H2 as worded is supported by both measures, but they answer different questions and only the first is what the p -value refers to.

Unlike H1, H2’s verdict does not depend on the choice of representation: the same selected-label cross-check described in Section 5.8.2 also rejects the null on both persona sides under the old, one-label treatment ($p_{\text{Holm}} = 0.0002$ for both). The model-side selected-label test survives aggregation to a single winner, whereas most of the human selected-label tests do not. This is not because of a smaller human raw change rate: at the label level, humans change their selected answer far more often than the model ensemble does (52.8% of human comparisons versus 18.4% of model comparisons, Table 5.5). The human tests are underpowered relative to the model tests, most plausibly because each human cell casts only three votes against the model ensemble’s much larger effective sample, not because the human effect is genuinely smaller.

Verdict: H2 is supported on both persona sides.

5.10 Is It the Persona, or Merely Having an Author?

Each comparison of a neutral to a persona condition carries a confound: persona prompts add an author line that neutral prompts lack entirely. A measured shift could reflect the persona’s content or merely the presence of an attribution. The `generic_human` condition separates them.

Table 5.9: Decomposition of the model framing effect using the generic-human control. Each row uses only the models that answered validly under all three of the neutral, generic-human and persona conditions for that side (a three-way intersection, stricter than matching models pairwise), so the two component distances and the confounded distance are computed over the same model set.

Comparison	Side a	Side b	95% CI (a / b)
Neutral → generic human	7.98%	8.01%	[7.23, 8.75] / [7.26, 8.77]
Generic human → specific persona	16.20%	19.30%	[14.14, 18.33] / [17.01, 21.61]
Neutral → persona (<i>confounded</i>)	16.73%	19.78%	[14.68, 18.89] / [17.50, 22.08]

Interpretation. A generic author line on its own produces a shift of almost exactly the same size on both sides (7.98%–8.01%). That is not negligible: naming any author at all changes how the models classify. The effect of the persona’s specific content, measured with the author line held constant, is roughly twice as large on both sides, and about as large as the full confounded comparison. These three distances describe what is associated with each step under matched models; they are not claimed to be additive components of one total effect.

The persona-content step is the larger of the two measured distances on both sides, while the author-line-alone step is smaller but still non-negligible. The approved proposal did not specify this decomposition, so the analysis is labelled secondary. Still, it turns a limitation that could otherwise only be acknowledged into one that is measured.

5.11 H3: Framing Effects and Ambiguity Type

Hypothesis. Framing effects will be stronger for author-relevant ambiguous texts than for author-independent ambiguous texts.

5.11.1 Raw Contrasts

Each text is reduced to a single number, its mean shift over the two persona sides, and the three categories are compared pairwise. The permutation null shuffles which category each text’s number belongs to; the effect size is the raw difference in mean shift, and its confidence interval is a text-level bootstrap.

Table 5.10: H3 pairwise contrasts on raw TV, Holm-corrected across the three contrasts.

Contrast	Difference	95% CI	p_{Holm}
Author-Independent vs Unambiguous	9.04pp	[2.63, 15.33]	0.0144
Author-Relevant vs Author-Independent	6.18pp	[0.30, 12.08]	0.0403
Author-Relevant vs Unambiguous	15.22pp	[9.08, 21.29]	0.0003

In raw terms, all three contrasts are significant after correction. However, the contrast H3 is actually about, author-relevant versus author-independent, is the **weakest of the three**: it has the smallest difference, the widest relative confidence interval and the largest corrected p -value. It is not a dramatic separation, and, as Section 5.11.2 shows next, this specific contrast does not survive being referenced against each text’s own sampling null, so “significant in raw terms” is the most this result supports on its own.

5.11.2 Referencing Against Each Text’s Own Sampling Null

The raw comparison is complicated by a property of small-sample TV. As set out in Section 4.6.5, TV between two three-vote samples is not zero under no effect, and it increases with the dispersion of the underlying distribution. The ambiguous categories

are by construction the dispersed ones. A raw category comparison is therefore a combination of baseline dispersion and the framing effect.

Table 5.11: Observed, expected-under-null, and excess TV by category (human, text-level).

Category	Observed TV	Expected under sampling null	Excess
Unambiguous	45.07%	43.07%	2.00pp
Author-Independent	54.10%	49.90%	4.20pp
Author-Relevant	60.28%	54.59%	5.70pp

The expected null TV is *itself* strongly ordered by category, ranging from roughly 43% to 55%. Once each text is referenced against its own null, the excess above that conditional-null expectation is on the order of two to six percentage points rather than forty-five to sixty.

Table 5.12: H3 contrasts on raw TV versus null-referenced excess TV.

Contrast	Raw	Null-referenced
Author-Relevant – Author-Independent	6.18pp, $p_{\text{Holm}} = 0.040$	1.49pp, $p_{\text{Holm}} = 0.697$
Author-Independent – Unambiguous	9.04pp, $p_{\text{Holm}} = 0.014$	2.21pp, $p_{\text{Holm}} = 0.697$
Author-Relevant – Unambiguous	15.22pp, $p_{\text{Holm}} = 0.0003$	3.70pp, $p_{\text{Holm}} = 0.287$
Holm-significant	3 of 3	0 of 3

Interpretation, stated carefully. The ordering is preserved in direction under null-referencing ($2.00 < 4.20 < 5.70$), but none of the three contrasts remains significant after correction.

Three separate conclusions follow, and they should be kept apart.

First, the raw contrasts show an observed difference of the kind the approved proposal anticipated. H3’s predicted ordering *is* present in observed shift magnitude.

Second, non-significance under null-referencing does not mean that the categories are equivalent. The excess-TV estimator subtracts an estimated conditional-null quantity, and the resulting point estimates are considerably smaller (1.49–3.70pp) than the raw estimates (6.18–15.22pp), while the confidence intervals are, if anything, slightly tighter in absolute width than the raw contrasts’ intervals (about 8.6–9.5pp wide versus 11.8–12.7pp raw). None of the three reaches significance because the estimated excess itself is small relative to that width, not because the interval has widened. Absence of evidence is not evidence of absence.

Third, excess TV measures observed distance above the chosen conditional-null expectation; it is a useful sensitivity statistic, not a validated decomposition into causal framing and baseline-noise components, since the pooled votes it refers to include observations from both conditions, and the spread of those votes could itself reflect condition differences. What it does show is that, given this reference, the category contrast becomes small and statistically inconclusive, so the raw ordering is sensitive to the choice of analytical reference, and this study cannot cleanly separate framing from baseline dispersion at 100 texts per category. These data therefore do not support a strong behavioural reading of the raw ordering.

This analysis was specified only after the raw result was known. It is therefore reported as a sensitivity analysis next to the raw contrasts, not in place of them; making it primary would amount to choosing an analysis by its outcome.

Verdict: H3 shows the predicted raw ordering, but the framing-specific contrast is inconclusive after referencing baseline vote dispersion. The point should be stated as an ordering in observed disagreement, not as a demonstrated framing-sensitivity gradient. Recall also from Section 3.4 that the categories were partly defined using an LLM judge’s own sensitivity to author-relevance, which makes H3 a transfer comparison rather than an independent discovery of that grouping principle.

5.12 H4: Human–Model Alignment

Hypothesis. The direction of framing-induced shifts in LLM classifications will partially align with human judgment shifts, but the magnitude of these shifts will differ.

Each component predicts something different, and either can hold without the other, so testing proceeds separately for each.

5.12.1 Magnitude

Two questions are easily confused here, and they must be kept apart.

Is there a covariation between shifts in humans and shifts in models? The 300 text-level means are used to compute Spearman’s $\rho = 0.331$, 95% CI [0.223, 0.428]; both persona sides are averaged on the human side, and both sides and six configurations are averaged on the model side (text bootstrap; Figure 5.5). The calculation is recorded in

`results/h4_text_level_covariation_verified.csv`. This positive relationship indicates nothing about the equality of the two magnitudes.

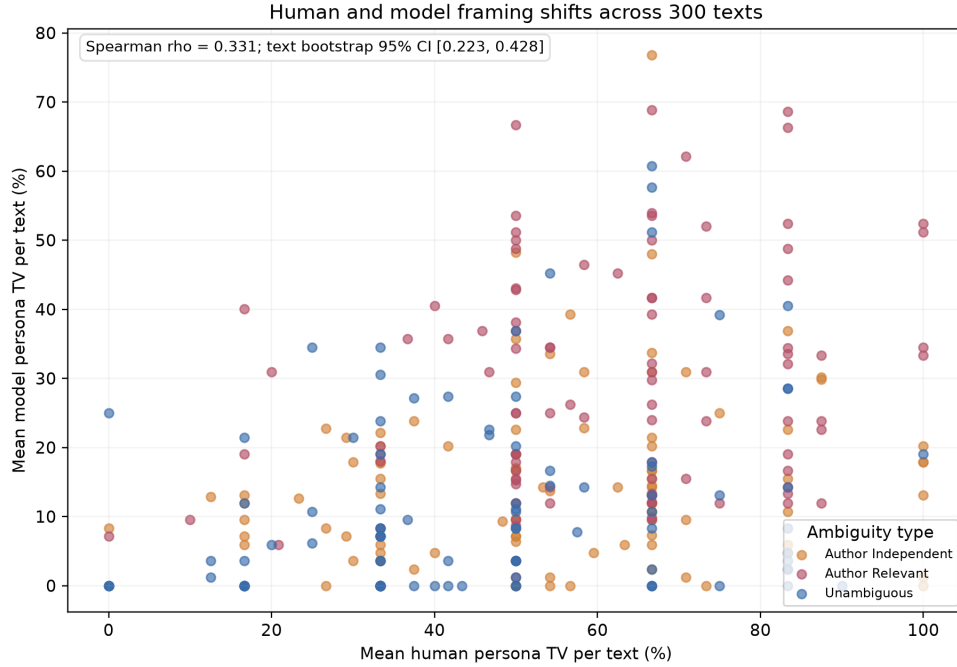


Figure 5.5: Mean human versus mean model shift magnitude for each of 300 texts, coloured by category.

Difference asks whether the sizes differ, and this is what H4 actually predicts. A paired Wilcoxon signed-rank test over 300 text-level pairs gives a mean paired difference of **34.86pp** (human minus model), 95% CI [32.23, 37.54], $p = 4.68 \times 10^{-48}$, with matched-pairs rank-biserial $r = 0.978$, close to the theoretical maximum, so that human shift is greater than model shift for almost all texts, not just on average (279 of 300 texts shift in that direction, 16 the other, 5 tied).

5.12.2 Is the Magnitude Gap an Artefact of Vote Counts?

Human vectors are built from three raters and model vectors from as many as seven models. Some of the difference might be arithmetic rather than behavioural: TV depends partly on the number of votes behind each vector. Rather than argue the point, this was tested directly.

The number of votes was fixed at three on both sides: human votes were subsampled to three per condition, and the *same three model identities* were drawn for the neutral and persona sides of each cell, holding ensemble composition fixed as well as size. Two hundred independent draws per comparison were performed, with the point estimate an equal-weight average across the balanced category $\times$ side $\times$ configuration strata.

Table 5.13: Magnitude gap before and after standardising both sides to three votes.

	Mean human – model gap
Unmatched	34.86pp
Count-standardised (3 votes vs the same 3 models)	$\approx$ 33pp
Reduction from count standardisation	$\approx$ 1.5pp (about 4% of the raw gap)

A note on precision. The standardised figure is a Monte Carlo estimate made with 200 draws per comparison, and is quoted deliberately as *approximately* 33 percentage points rather than to two decimals.

When the same inputs and the same seed are used, the same estimate is obtained to six decimal places on repeated runs. The *ordering* of votes within a cell is what makes it move slightly. Subsampling draws indices, so if a condition has more than three votes, the same multiset presented in a different order yields a different sampled subset under the same seed. Re-ordering the votes within each cell, sorting or reversing them while leaving the multiset untouched, moves the estimate between 33.34 and 33.37pp, and five independent seeds span 33.32 to 33.37pp.

That range, roughly 0.04pp either side, is about where a one-decimal figure rounds up or down, so no decimal is claimed. This is not crucial to the interpretation: count standardisation reduces the estimated gap by about 1.5 percentage points for all orderings and all seeds tested.

Interpretation. Standardising both sides to the same number of votes cast reduces the estimated gap from 34.86 to approximately 33 percentage points, a decrease of about 1.5 points (4%), and the gap remains within each of the categories. In line with the precision noted above, this is expressed to the nearest whole point, not to two decimal places, since the second decimal place moves by more than that between orderings and seeds. This procedure is limited in that it does not causally identify the proportion of the gap explained by vote counts, and it does not remove the between-person/within-model design asymmetry discussed below: humans still shift substantially more than the model ensemble, and this is not an artefact of counting.

Count standardisation leaves two qualifications untouched. The human comparison is **between-person** (different raters in each condition) and the model comparison is **within-model**, with the same model answering both conditions. Part of the human change is thus between-rater variability for which there is no model counterpart. And each model produced only one generation per condition, so model-side stochastic variability is unmeasured.

5.12.3 Direction

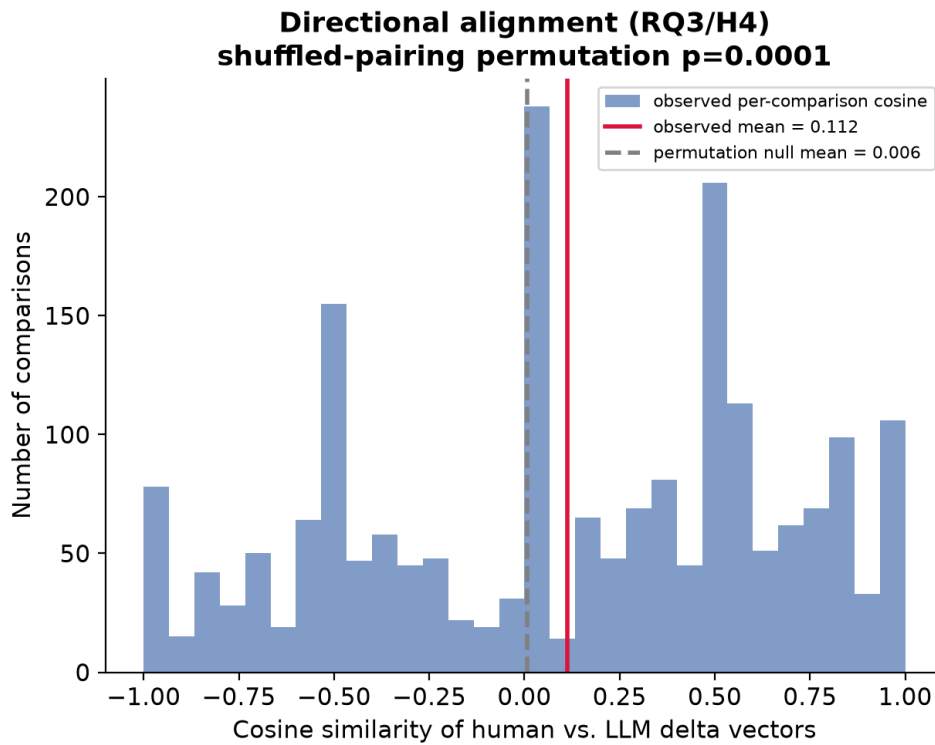


Figure 5.6: Distribution of per-comparison directional cosine between human and model delta vectors, one point per (text, side, prompt-configuration) comparison, with the observed mean and the chance-pairing null marked.

Mean cosine between human and model delta vectors is **0.112**, 95% CI [0.072, 0.152], against a shuffled-pairing permutation null (mean 0.006), $p = 0.0001$. The direction of the model ensemble shift is thus more closely associated with the direction of the corresponding human shift for the text than random pairing would produce, but a cosine of about 0.11 is a weak, partial association, not a close match.

Coverage. This is calculated at the individual comparison (per text, per side, per prompt-configuration) level throughout, not as an average over a coarser, per-text level. Of the 3,600 total such comparisons, **2,020 (56%)** are defined, and 252 of the 300 texts (84%) have at least one defined comparison among their set of twelve. Comparisons with no direction to compare are called undefined and are not treated as zero.

Verdict: H4 is partially supported. The magnitude component is **supported**: the magnitudes differ, substantially and robustly to count standardisation, as H4 predicts. The direction component is **partially supported**: alignment is above chance but weak, which is the literal meaning of H4’s word “partially”.

The direction analysis is not immune to small-sample noise either. Cosine is invariant to multiplying a vector by a positive scalar, but drawing three votes rather than seven is not scalar multiplication: it can change a delta’s direction, and it can make the delta exactly zero, in which case direction becomes undefined rather than being preserved.

5.13 Where the Probability Mass Moved

The biggest descriptive shifts are **+4.0pp** for *guilt* in the author-independent category and **−5.4pp** for *sadness* in the unambiguous category, with gains and losses elsewhere for *fear* and *anger*.

The result is different if the test is extended to the model ensemble. The original analysis covered only the 33 (category × emotion) human contrasts. All 66 (source × category × emotion) tests were repeated using the identical sign-flip permutation test and corrected in a single pass using the Benjamini–Hochberg FDR method (Table 5.14), yielding **7 of 66** significant at $q < 0.05$. One of the seven is human: the +4.0pp gain in *guilt* on author-independent texts is significant at this combined scale ($p_{\text{BH}} = 0.041$), but not on the human-only family tested alone. The other six are model effects.

Table 5.14: Per-emotion shifts significant under BH-FDR at $q < 0.05$ (of 66 tests).

Source	Category	Emotion (mean change)	p_{BH}
Human	Author-Independent	guilt +4.0pp	0.041
LLM	Unambiguous	disgust −0.7pp	0.027
LLM	Unambiguous	pride +1.7pp	0.009
LLM	Author-Independent	joy −1.9pp	0.012
LLM	Author-Independent	pride +1.9pp	0.009
LLM	Author-Relevant	joy −5.1pp	0.009
LLM	Author-Relevant	relief +3.6pp	0.012

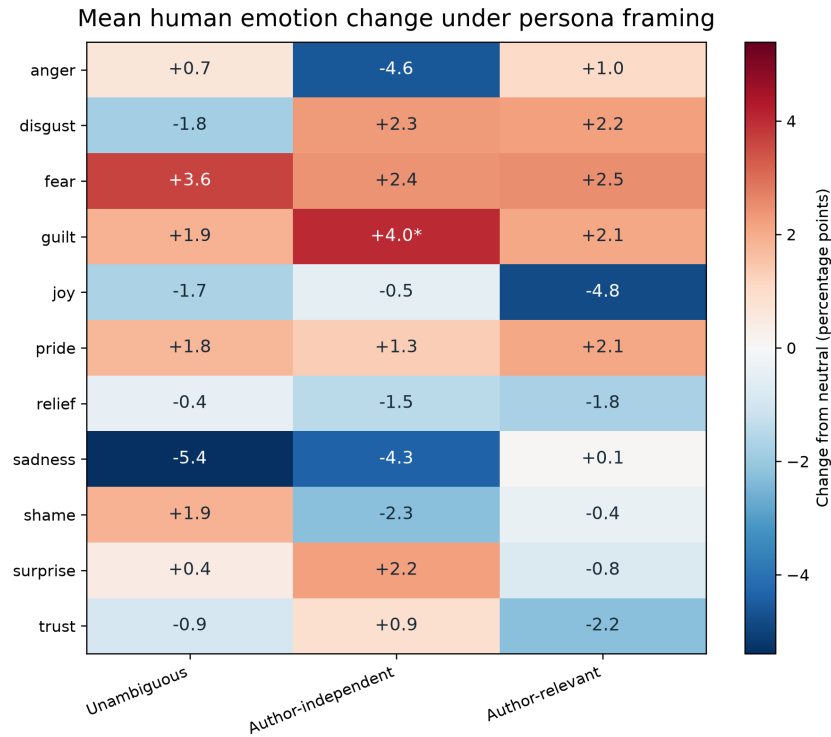


Figure 5.7: Mean signed change in probability mass per emotion and category (human, text-level, percentage points).

Interpretation. Of the 33 human cells, the only one to survive correction is the movement of guilt, already noted above as one of the largest descriptive shifts (guilt, author-independent), rather than the general “self-conscious emotions gain, sadness and anger lose” pattern visible in the heatmap; the other 32 human cells, including the numerically larger sadness and fear movements on unambiguous texts, do not survive correction. The six model effects are not random noise but form a coherent pattern: on both author-independent and author-relevant texts, the ensemble shifts away from *joy* and towards a more specific positive emotion (away from *joy* towards *pride*, and away from *joy* towards *relief*), while on unambiguous texts the ensemble shifts towards *pride* and away from *disgust*. Given a persona, then, the model ensemble tends to swap a generic positive emotion for a more specific one rather than bring in threat-related emotions. The result is no longer reported as a uniform negative finding. It is a small, mostly model-side, directionally coherent effect, sitting on top of a human per-emotion pattern that this sample size leaves largely unconfirmed.

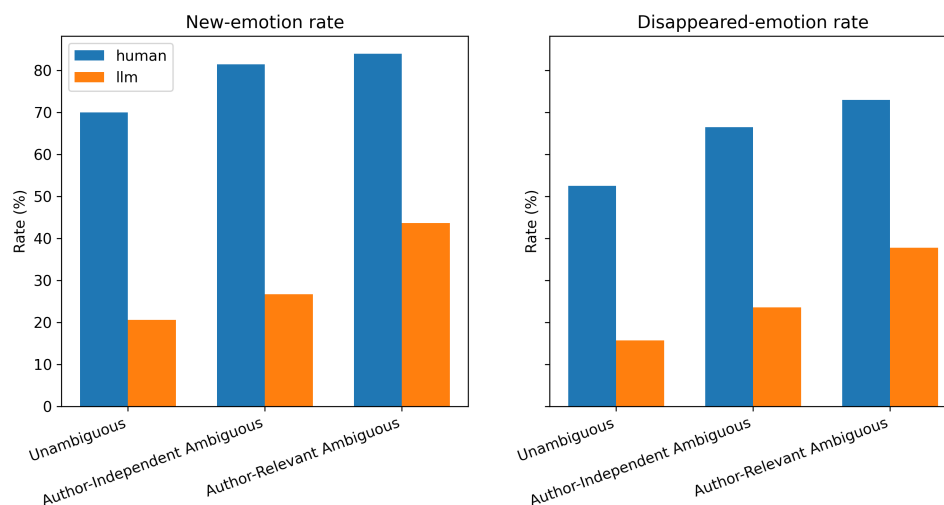


Figure 5.8: Rates at which emotions appear or disappear between conditions, by source and category.

The frequency at which an emotion appears post-framing that was absent beforehand rises with category for both sources, but is far higher for humans (70.0%, 81.5%, 84.0%) than for the model ensemble (20.3%, 26.3%, 43.4%). Disappearance rates follow the same ordering (humans 52.5%, 66.5%, 73.0%; models 15.7%, 23.1%, 37.3%). What is being observed here is a change in the *support*, the set of emotions receiving any vote at all, and not a transition tied to any specific rater or model; because the human and persona conditions were rated by disjoint people, no individual vote is ever observed to change. The difference is mostly one of structure, not behaviour: with only three votes, one human vote landing on a previously unvoted emotion is enough to change the support, whereas a seven-model ensemble requires the same event across a larger denominator, so support detection is itself a function of sample size as well as of the underlying vote distribution.

5.14 A Traceable Strong-Shift and Stable-Condition Example

Text 2105 is about passing exams; the original emotion word has been removed from the stimulus. Its neutral human votes are *pride*, *joy*, *joy*. The four independent human votes for Persona A (a student who had struggled all year and barely managed to pass) are *sadness*, *relief*, *relief*, *relief*. The two vote sets have disjoint emotion supports, so the TV

distance is 1.0, a 100% distributional shift. Persona B is a high-performing student who expected to pass: its votes are *pride*, *joy*, *joy*, exactly the neutral distribution, giving TV distance 0. The case was picked for its observed shift pattern, and the juxtaposition does not necessarily imply that the persona caused a specific individual to change their answer. Source: `data/human_annotations.csv` (text 2105), with persona wording in `data/sampled_texts.csv`.

For the zero-shot, numbered-list configuration, the OpenAI model outputs *pride* for neutral framing, *relief* for persona A, and *pride* for persona B. The recorded explanations are, respectively, accomplishment, relief after a difficult year, and success in line with high expectations (`data/experiment_results_all.csv`, text 2105). These responses indicate that the explanation field was gathered and that this particular model/configuration is qualitatively similar to the human contrast, but not that these explanations are a mechanism, and not that every model/configuration behaves this way.

5.15 Does Framing Concentrate or Disperse Interpretation?

Magnitude and dispersion are different things. A distribution can move a considerable distance and still end up with one firm new response, or it can spread thin with little movement. At the text level, the two are empirically close to unrelated in both sources: for humans $\rho = 0.092$, $p = 0.113$; for the model ensemble $\rho = -0.048$, $p = 0.406$.

A per-text mean entropy change is calculated for each cell and tested with a one-sample Wilcoxon signed-rank test against a null value of zero (since each text contributes exactly one value, this avoids the pseudoreplication a per-comparison test would introduce); Benjamini–Hochberg correction is applied across all six tests.

Table 5.15: Entropy change under framing (text-level, BH-FDR corrected across six tests).

Source	Category	Δ entropy	p_{BH}	Significant
Human	Unambiguous	+0.198	0.0002	yes
Human	Author-Independent	+0.171	0.0002	yes
Human	Author-Relevant	+0.161	0.0030	yes
Model	Unambiguous	+0.028	0.226	no
Model	Author-Independent	+0.012	0.230	no
Model	Author-Relevant	+0.021	0.350	no

Interpretation. Humans reliably agree *less* among themselves under persona framing, in all three categories. No dispersion effect is detected on the model ensemble in any

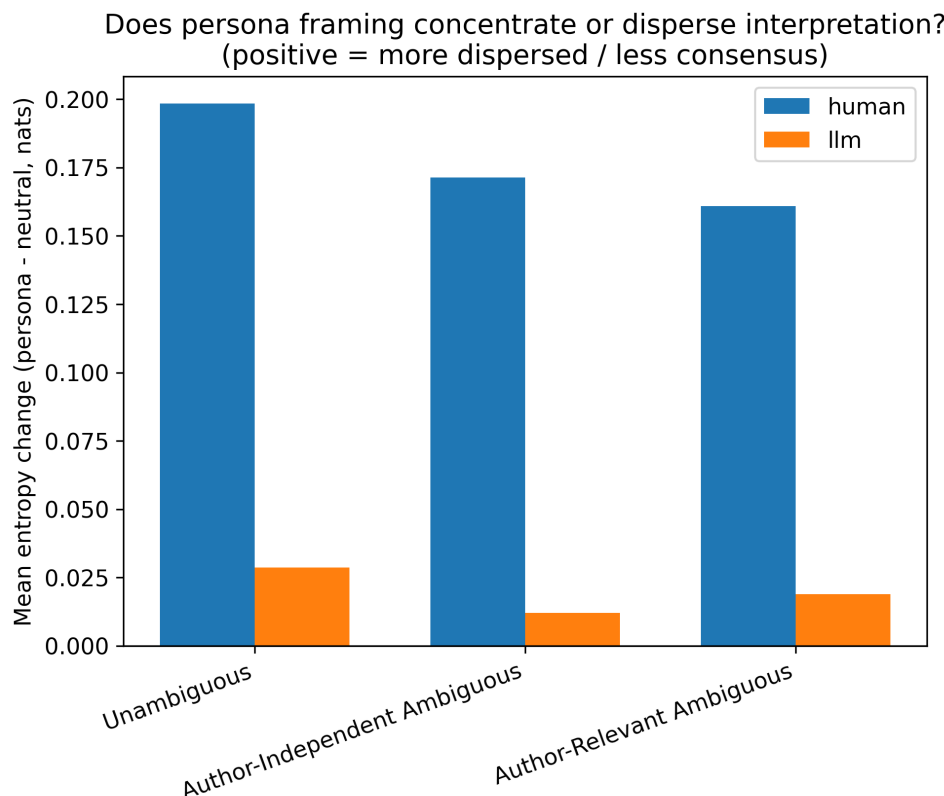


Figure 5.9: Mean entropy change from neutral to persona framing, by source and category. Positive values indicate greater dispersion.

category: it moves its probability mass (Section 5.9) without detectably changing its internal agreement. These are two separate sets of tests, one significant throughout and one not. A significant result next to a non-significant one does not by itself show that there is a statistical difference between the two sources, and that difference was not tested directly. Put cautiously, human dispersion increases were detected, but model dispersion increases were not. This hints at an asymmetry that no magnitude-only measure would reveal, but it stops short of a demonstrated human–model contrast.

A caveat parallels the H4 count-standardisation check (Section 5.12.2): every neutral human cell rests on exactly three votes, whereas persona cells rest on three to seven, and plug-in entropy rises mechanically with the number of votes, so part of the raw human entropy change above is a vote-count artefact rather than framing. Standardising both conditions to three votes each (`results/count_standardisation_summary.csv`) reduces the mean human entropy change from 0.177 to 0.135 (95% CI [0.089, 0.181]),

roughly a quarter smaller, but it remains clearly positive; the model change is not distinguishable from zero after standardisation either (0.013, CI $[-0.003, 0.029]$).

5.16 How Often Does Framing Change Nothing?

In 8.5% of human comparisons and 40.2% of model comparisons the persona changed **nothing at all** (TV exactly zero). The median TV for humans is 60.0% and for the ensemble is 14.3%.

Interpretation. This is not because every cell moves by a moderate amount: the mean shift of the model ensemble is 18.3%. A large minority does not move at all, and the rest move considerably more. This also explains why so many of the cosines in Section 5.12.3 are undefined: a zero delta has no direction. Part of the human–model asymmetry is again structural, since a seven-way vote has more available states than a three-way vote.

5.17 The “Unchanged” Category Conceals Movement

A conventional dominant-label comparison records a change only when the winning emotion flips. Broken into the five outcomes defined in Section 4.4.3:

Table 5.16: Dominant-set outcomes. These set relationships differ from changes in an alphabetically selected single winner.

Outcome	Human	Model ensemble
Identical set	34.3%	78.9%
Single emotion split into a tie	17.0%	2.6%
Tie resolved to a single emotion	8.2%	3.4%
Partial overlap of two ties	7.0%	0.4%
Fully disjoint top sets	33.5%	14.8%

Interpretation. In 33.5% of the comparisons, human top sets are completely disjoint and 34.3% are identical, while the remaining 32.2% involve a tie forming, resolving, or partially overlapping. The corresponding figures for the model ensemble are 14.8%, 78.9%, and 6.3%. These are *set* relationships, not the changed/unchanged decisions made by the selected-label method. That method records 52.8% of human and 18.4% of model comparisons as changed (Section 5.6); an overlapping top set can still select a different winner.

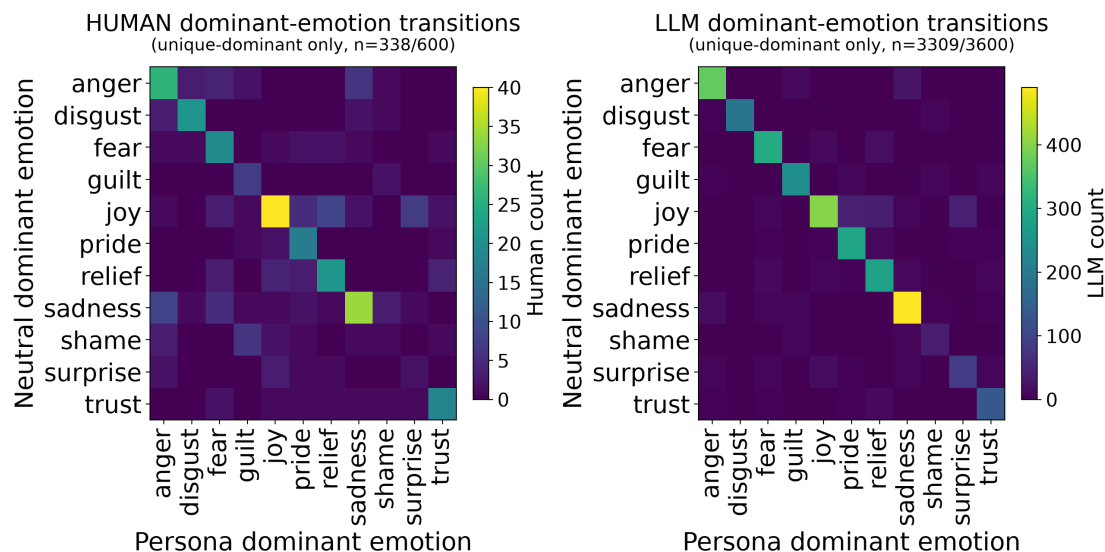


Figure 5.10: Dominant-emotion transition matrices for human and model judgments, restricted to comparisons where both sides have a unique dominant emotion.

5.18 Prompt Structure Has No Measurable Effect

A repeated-measures Friedman test across the six prompt configurations (text-level, $n = 300$) gives $\chi^2 = 7.298$, $p = 0.199$. A test of the selected-label change rate, rather than the shift, gives $\chi^2 = 1.709$, $p = 0.888$; neither representation detects a prompt-configuration effect.

Interpretation. There is no measurable effect on the size of the model framing effect in this dataset when prompt design (zero-, one- or few-shot) is crossed with either numbered or inline label presentation. Non-rejection does not establish that prompt configurations are equivalent. Since the approved proposal contains no numbered hypothesis about prompt structure, this is a supplementary observation and should not be cited as a test of a formal hypothesis. Its practical use is as a robustness check: the H2 effect is detectable when the six configurations are combined, and descriptive effects are present across configurations, but a configuration-specific equivalence claim is not supported.

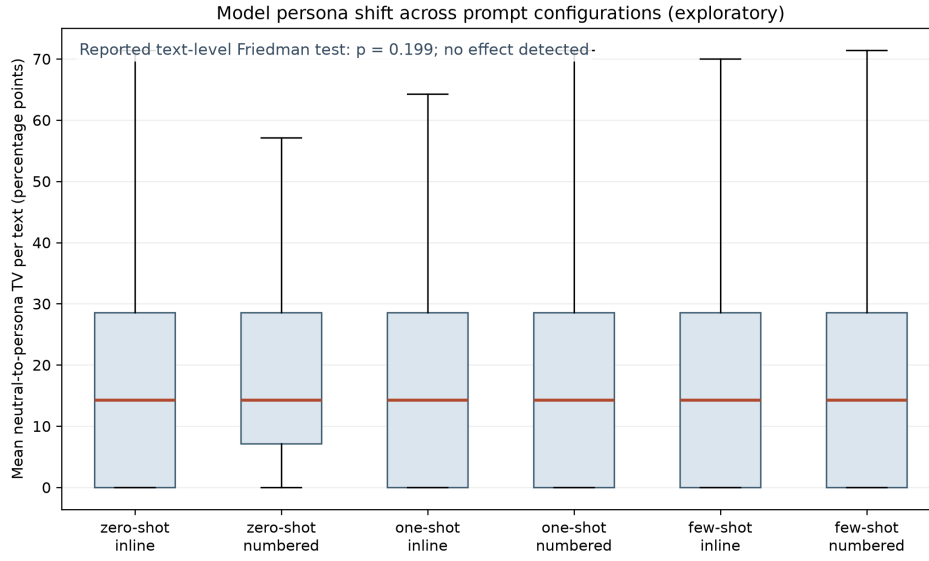


Figure 5.11: Per-text shift magnitude across the six prompt configurations.

5.19 Supplementary Check: Human Rating Agreement

The proposal also requested a check on how well human readers agree with one another. Fleiss’ κ asks: *how much more often do readers agree than we would expect based on the prevalence of each emotion label?* A value of zero is the level of agreement one might expect from those label frequencies; a larger value indicates greater agreement. It serves only as supporting description; it is not another hypothesis, nor the main reason for using emotion vectors. It is calculated for the 754 of 900 text-by-framing cells that have exactly three human ratings each, separately for each sampled category and framing. All neutral cells have three ratings, whereas some persona cells have four to seven and are omitted from this particular statistic. Table 5.17 gives both kappa and the number of included texts in parentheses (source: `results/human_interannotator_agreement_verified.csv`).

Table 5.17: Human Fleiss’ κ by sampled category and framing; included cells in parentheses.

Category	Neutral	Persona A	Persona B
Unambiguous	0.465 (100)	0.270 (78)	0.350 (76)
Author-Independent Ambiguous	0.344 (100)	0.229 (72)	0.250 (79)
Author-Relevant Ambiguous	0.294 (100)	0.177 (75)	0.159 (74)

Neutral agreement is highest in the unambiguous pool and lowest in the author-relevant pool. The persona columns indicate only the exactly-three-vote subset, with different texts included across the different framings. They therefore do not establish any causal change in agreement, nor do they replace the within-text framing analysis above.

5.20 Summary of Hypothesis Outcomes

Table 5.18: Summary of hypothesis outcomes.

Hypothesis	Key evidence	Verdict
H1	Author-relevant significant both sides ($p_{\text{Holm}} = 0.0066, 0.0112$); unambiguous non-significant; author-independent A only, direct A–B inconclusive; selected-label tests: 1 of 6 significant under reverse-alphabetical tie-break, 0 of 6 under alphabetical	Partially supported
H2	Paired test $p_{\text{Holm}} = 0.0002$ both sides; also significant under the selected-label cross-check; 21.3% of model pairs changed label; generic-author control shows movement for both neutral-to-generic and generic-to-specific steps (not additive components of one total effect)	Supported
H3	Raw: 3 of 3 contrasts Holm-significant. Null-referenced: 0 of 3	Predicted raw ordering observed; framing-specific contrast inconclusive
H4 magnitude	34.86pp difference, surviving count standardisation ($\approx 34.9 \rightarrow \approx 33\text{pp}$); positive text-level covariation (Figure 5.5)	Supported (magnitudes differ)
H4 direction	Mean cosine 0.112, $p = 0.0001$ vs chance pairing	Partially supported
Per-emotion (exploratory)	7 of 66 (source $\times$ category $\times$ emotion) tests significant under BH-FDR: 1 human (guilt), 6 model	Small, mostly model-side effect
Prompt structure	Friedman $p = 0.199$ (vector), $p = 0.888$ (selected-label)	No effect detected (exploratory)

Hypothesis outcomes

Primary tests and key sensitivity results

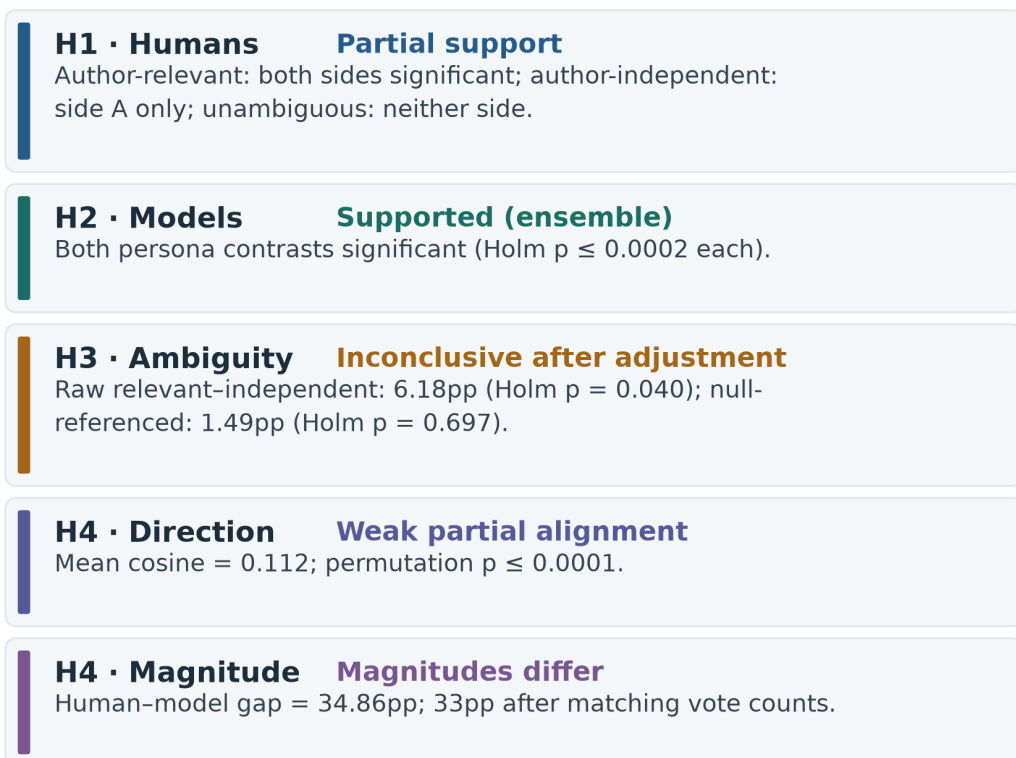


Figure 5.12: Overview of hypothesis outcomes across the analysis.

6 Discussion

This chapter interprets the results with reference to the three research questions. Results that cut across the individual hypotheses are drawn out as stand-alone findings, and the chapter closes with the limitations of the design and what future research might address.

6.1 Interpretation with Respect to the Research Questions

6.1.1 RQ1: Does Perceived Authorship Influence Human Emotion Judgments?

The extent to which a framing effect is found varies by category.

Human judgments shift significantly under author framing for author-relevant ambiguous texts on both persona sides, but not for unambiguous control texts on either side. This pattern is in line with H1. The controls' non-rejection does not prove a zero effect, or a separation of the persona's content from the other alterations the author cue makes.

The author-independent category is significant under persona A and not under persona B. A split decision of this kind at the significance threshold cannot on its own show that the personas have different effects. The exploratory direct comparison of their text-level raw TV shifts is 4.44 percentage points, with a 95% bootstrap interval from -2.08 to 11.06 points (Section 5.8). Because this interval includes zero, these data do not establish an asymmetry between A and B. Whether the category responds to incidental persona content remains a question for further investigation, not a demonstrated result.

The size of these effects requires a qualification, developed in Section 5.11.2. Taking the vote distribution of each text alone, sampling noise from three votes would account for a substantial portion of the raw human shift. The effect is detected in the sense that it exceeds the permutation null, but the raw TV magnitude is a crude figure that

reflects not only the effect of framing but also ordinary sampling noise, so it should not be interpreted as a pure measure of framing’s own contribution to the movement.

The second qualification concerns how the effect was detected in the first place. Repeating the identical permutation test on the selected label rather than the vote vector rejects the null in one of six cases under a reverse-alphabetical tie-break convention, and none under the alphabetical convention used throughout this thesis (only these two conventions were tested; Section 5.8.2). Thus, nearly all of the support for RQ1 comes from the distributional representation; under the majority-label representation, one of six tests would have found a significant effect, and under this thesis’s own convention, none. The gap is largely an artefact of the representation. Its exact size, however, is not intrinsic to the data but is determined by an otherwise-arbitrary tie-breaking decision.

6.1.2 RQ2: Does Perceived Authorship Influence LLM Classifications?

Yes, and with fewer qualifications than RQ1 requires.

The ensemble effect is significant on both persona sides. The observed label-change rates are nonzero for all seven models, although per-model significance tests were not performed. The exploratory test detects no prompt-configuration effect, but nor does it demonstrate equivalence across formats. The differences observed across models and formats make the overall result less dependent on any particular setting, subject to the single-generation limitation below. This extends the finding of Santurkar et al. 2023 that persona conditioning shifts model outputs, from opinion elicitation to emotion classification, and matches the direction of Zhang et al. 2018 on persona-conditioned dialogue.

The generic-human control makes the claim considerably sharper. The neutral-to-generic-author contrast is approximately 8% on both sides (7.98% and 8.01%), while the generic-author-to-specific-persona contrast is 16.2% and 19.3%. These distances are sensitive both to the presence of an attribution and to its specific content, but they are not additive components of a total effect, so they cannot establish what proportion of the overall shift is due to identity specifically. Without this control, the two could not have been distinguished at all.

The claim has two limits. First, the significance test identifies *coordinated* redistribution across models; 61.2% of the cells contribute no variance to this test’s null, and a scenario in which every cell contained exactly one dissenting model would yield $p = 1$ even though

every cell changed. The complementary raw change rate, 21.3% of model pairs altering their label, is reported because the two measures answer different questions. Second, each model produced one generation per condition, so no distinction is made here between a framing response and ordinary stochastic variation in generation. That would require multiple generations under the same prompts, which were not gathered.

6.1.3 RQ3: Do LLMs Resemble Humans in Direction and Magnitude?

Here the answer splits cleanly in two, and that split is the most substantive finding of the thesis.

In direction, models resemble humans weakly but reliably. Model delta vectors, when compared with human ones, have a mean cosine of 0.112, well above a shuffled-pairing null. Models do not merely shift when framing is applied; they shift in a direction associated with the direction the same text moves for human readers. A cosine of 0.11 is nonetheless a small association: the resemblance is genuine but incomplete, as H4 suggested. The criterion is also more stringent than the usual label-agreement comparison. It does not ask whether the model reaches the same answer as a human, but whether it *responds to context* in the same direction, which is closer to the question posed by ambiguity-aware approaches such as Hong et al. 2025.

In magnitude, models and humans differ substantially. Humans shift more, by roughly 34 percentage points, and this survives standardising both sides to an equal number of votes: doing so reduces the estimated gap by only about 1.5 points (roughly 4%), and it does not remove the between-person/within-model design asymmetry discussed below.

One further pattern escapes magnitude altogether, although it calls for caution. Under framing, humans become measurably *less in agreement with one another*: entropy rises significantly in all three categories. No dispersion effect is detected for the model ensemble in any category. These are separate tests, and a significant result on one side alongside a non-significant result on the other does not by itself establish that humans and models differ from each other statistically; that difference was not tested directly. Taken as two separate results rather than a demonstrated contrast, the pattern suggests that models redirect their probability mass while human judgments also become more dispersed. An analysis reporting shift magnitude alone would never see this.

For the models, then, the results indicate sensitivity to author framing, weak directional alignment with human judgments, and smaller observed shifts. Human entropy increases were statistically detected, but not model entropy increases; whether those entropy responses differ between humans and models remains untested.

6.2 Cross-Cutting Findings

6.2.1 What Aggregation Costs

This finding is a measurement rather than an argument, and it is the clearer of the two methodological results in this section. A selected-label analysis of these same data would have recorded 38.7% of human comparisons and 41.4% of model comparisons as showing no change, when the underlying distribution demonstrably moved (under the alphabetical tie-break convention; crediting the label method for tracking the complete set of tied top emotions instead lowers these to 25.8% and 38.7% respectively). For the model ensemble, the comparisons such an analysis cannot see outnumber those it detects by more than two to one, and for humans the invisible comparisons are nearly three-quarters as numerous as the detected changes (38.7% invisible against 52.8% detected).

Label-based analysis is not always inappropriate. For the question asked here, however, whether framing shifts interpretation, the choice of representation decides whether the phenomenon can be observed at all. The human side shows this most clearly: a study designed identically but analysed on majority labels would have detected a statistically significant human framing effect in one of the six H1 tests (reverse-alphabetical tie-break convention) or none at all (this thesis’s alphabetical convention) rather than three, and would most plausibly have been interpreted as providing little or no evidence of human sensitivity to author framing. The model-side selected-label test, by contrast, still detects a significant effect under either representation (Section 5.9.1), so for the model ensemble the choice of representation changes how much of the shift is visible, not whether an effect is detected at all.

This concern was raised in principle by Aroyo and Welty 2015 and Basile 2020; the figures above put a size on it. They are also consistent with Pavlick and Kwiatkowski’s 2019 argument that disagreement in subjective annotation is frequently inherent rather than erroneous. That literature argues that aggregation loses signal; this measurement shows how much is lost and of what kind.

The finding on baseline agreement points the same way. Overall, the human and model dominant labels agree for only 58.0% of comparisons under neutral framing (as high as 74.2% on unambiguous texts, and as low as about 50% on the two ambiguous categories), but within both the matching and non-matching buckets, full-vector similarity spans almost the entire possible range. Reporting only the binary match rate compresses a continuous and highly variable quantity into a single number that conceals its own variance, and even that single number shows the two sides starting from a noticeably different reading of the ambiguous texts before any persona is introduced.

6.2.2 Small Samples Make TV Deceptive

One finding came out of methodological scrutiny rather than the original design, and it applies well beyond this thesis.

Total variation distance between two small vote samples is not zero when no effect is present, and it grows with the dispersion of the underlying distribution. Two independent three-vote draws from the same distribution routinely disagree, and they disagree more when that distribution is spread out. Since categories defined by ambiguity are, by construction, categories defined by dispersion, comparing raw TV across such categories confounds the quantity of interest with the measurement's own noise floor.

Referencing each text against its own conditional sampling null showed that expected null TV is itself ordered by category, ranging from roughly 43% to 55%. The excess above that conditional-null expectation is on the order of two to six percentage points, not forty-five to sixty. Any study that compares distributional distance across groups differing in baseline dispersion runs into this problem, and it stays hidden unless the null is constructed explicitly.

6.3 Limitations

These results are constrained by the limitations set out below. They have been gathered here rather than scattered throughout the results chapter, so that they can be weighed together.

Vote counts are small and unequal between sides. Human vectors rest on three raters per condition and model vectors on up to seven. This gives rise to several struc-

tural rather than behavioural human–model asymmetries reported in Chapter 5 (tie frequency, new-emotion rates, zero-shift rates, dominant-set stability). Count standardisation (Section 5.12.2) addresses the magnitude comparison specifically, but not these descriptive asymmetries.

The human comparison is between-person; the model comparison is within-model. Different people rated each framing condition, whereas the same model answered both. Part of the measured human shift is between-rater variability with no model counterpart. The problem is built into the design, and matching vote counts does not fix it. It is the main caveat on the H4 magnitude result.

H3’s categories were partly defined by the property H3 tests. Texts were assigned to the author-relevant group using an LLM judge’s own sensitivity to whether author framing flips the reading. H3 is therefore a transfer comparison: whether a grouping derived from that judge’s sensitivity predicts effects in newly collected data, and not an independent discovery that author-relevance is the operative principle. This is a sharper limitation for the model-side ordering specifically: the model ensemble’s framing effect is reported as ordered by the same category that the LLM was used to define, so part of that ordering is true by construction rather than an independent finding.

The category ordering is not separable from baseline dispersion. H3’s contrasts are significant in raw TV, but not significant once referenced against each text’s sampling null. This is not because the null-referenced intervals are wider; they are, if anything, narrower in absolute width than the raw contrasts’ intervals. Rather, the estimated excess over the conditional null shrinks much more than the interval does, so the same interval now spans zero. This does not demonstrate equivalence between categories, but it does show that this design cannot attribute the raw ordering to framing rather than to baseline dispersion at 100 texts per category.

Model stochasticity is unmeasured. One generation per condition means the model figures are single draws, not stable per-model expectations.

No generic-human condition was collected for humans. The decomposition separating author presence from persona content is available for models only, so the equivalent human question remains open.

Per-emotion effects are underpowered. Only 7 of the 66 per-emotion tests survive FDR correction, and 32 of the 33 human tests do not, including the numerically largest

movements in the heatmap. The broader descriptive pattern may be real, but this study establishes only the one human effect and the six model effects that survived correction, not the pattern as a whole.

Cross-text respondent dependence cannot be modelled, and within-text independence is not fully guaranteed either. No collection form repeated a text across both of its conditions, which is a property of the form design and can be verified from the assignment records. Respondent identifiers were not retained, however, and there is no way to determine whether the same individual responded to both conditions of the same text; that is, the same individual could in principle have rated both conditions without this being evident in the retained data. The same lack of participant tracking means cross-text correlation, one person’s answers on one text being linked to their answers on another, also cannot be ruled out or quantified: each form carried on the order of thirty items, and a one-to-one correspondence between forms and participants would bound the possible correlation structure, but no such correspondence is verifiable from what was retained. The within-text permutation null also requires comparable respondent groups under the null; disjoint form assignments narrow this risk but do not, on their own, establish that exchangeability assumption.

6.4 Future Directions

The limitations above suggest five extensions.

Collect human annotations at matched vote counts. If human raters per condition were raised to seven, the main structural asymmetry would be removed, and the descriptive comparisons (tie rates, new-emotion rates, dispersion) could be interpreted behaviourally rather than arithmetically.

Add the generic-human condition to the human study. This would permit the same neutral-to-generic-author versus generic-to-specific-persona comparison already run on the model side, so that the two separately measured distances could be compared directly for humans instead of assumed from the model pattern.

Sample repeated generations per model. Several generations per prompt would separate framing response from generation variance and convert the model figures from single draws into estimates with their own uncertainty.

Define ambiguity categories independently of the tested property. If texts were assigned to categories on criteria not derived from an LLM judge’s own framing sensitivity, H3 would become a direct test rather than a transfer comparison.

Retain respondent identifiers. Anonymised respondent identifiers linked to text and condition would allow a participant-aware analysis and remove the one dependence structure this study cannot quantify.

Another direction comes from the entropy finding rather than from a limitation. A significant human dispersion increase was detected and no model dispersion increase was. The apparent contrast was not tested directly, and to the author’s knowledge the existing literature on persona prompting has not examined it. It relates to work modelling annotators individually rather than in aggregate [Davani, Díaz, and Prabhakaran 2022; Uma et al. 2021]: if human interpretive uncertainty is itself structured, a model that reproduces answers without reproducing uncertainty is capturing only part of the behaviour. Can models be made to show human-like interpretive uncertainty, and not just human-like answers? That question looks both open and tractable.

7 Conclusion

This thesis explored whether perceived author identity measurably affects the interpretation of emotion in ambiguous text, for both humans and large language models, and whether the two respond in comparable ways. Author attribution was manipulated experimentally while the textual content was held constant. In a first pass, one emotion was selected per condition and the hypotheses were tested using that label. A second pass kept all eleven emotion proportions, retested the hypotheses, and identified changes that the first pass had missed.

7.1 Summary of Findings

Author framing shifts human emotion judgments for author-relevant ambiguous texts, on both persona sides, and has no statistically detectable effect on unambiguous control texts. The second result fits H1, but it cannot confirm a zero effect or show that the manipulation acted only on author information. In the author-independent category one persona contrast is significant and the other is not; a direct comparison does not establish different effect sizes.

The model ensemble is sensitive to author framing on both persona sides. All seven models have nonzero observed label-change rates, although separate per-model significance tests were not run. No prompt-configuration effect was detected, so equivalent effects across formats cannot be claimed. A previously unanalysed generic-human control separates this into two steps under matched models: adding an unspecified author line alone moves classifications by about 8% on both sides, and the step from there to a specific persona moves them by 16–19%. These are two separately measured distances, not additive components of one total effect.

The predicted ordering across ambiguity categories is present in raw shift magnitude, and all three pairwise contrasts survive correction. Referencing each text against its own conditional sampling null, however, shows that the shift expected under no effect is

itself strongly ordered by category. The excess above that conditional-null expectation is small enough that none of the contrasts remains significant. The raw ordering agrees with H3’s prediction, but evidence for a greater *framing-specific* effect in the author-relevant category is inconclusive at this sample size.

On human–model alignment, the two components of H4 separate cleanly. Models track the *direction* of human shifts better than chance, but only weakly, which is what H4’s word “partially” predicts. Humans show much larger shifts in *magnitude*, by roughly 34 percentage points, and this survives standardising both sides to equal vote counts. A further pattern, beyond magnitude, was not anticipated by the original hypotheses. It is reported cautiously, since the two sides were not tested against each other directly: under framing, human judgments become measurably less in agreement with one another, while the model ensemble shows no statistically detectable dispersion effect in any category. Read as separate results rather than a demonstrated contrast, they suggest that models redirect probability mass while human group judgments also become more dispersed.

Two findings are reported as negative or near-negative results. Under multiple-comparison correction, almost none of the individual per-emotion shifts remain significant: of 66 tests spanning both sources, only one human effect (a gain in guilt on author-independent texts) and six model effects survive, and the model effects form a coherent pattern of moving away from *joy* towards more specific positive emotions (*pride*, *relief*) rather than towards threat-related ones. The exploratory comparison found no effect of prompt configuration on framing-effect magnitude. One further result was unplanned: most of H1’s support depends on the choice of representation. The identical permutation test applied to the selected label rather than the vote vector rejects the null in one of its six cases under a reverse-alphabetical tie-break convention, and none under the alphabetical convention used throughout this thesis (only these two otherwise-arbitrary conventions were tested). A majority-label version of this same study would have detected a statistically significant human framing effect in one of those six tests at best, or none under this thesis’s own convention.

7.2 Contribution

Substantively, the thesis offers evidence of a real and measurable influence of perceived authorship on the interpretation of emotion by both humans and language models. It

adds a control condition that isolates persona content from author presence, and it compares the two sides directly under one design.

The methodological contribution may reach further still. Whether judgments are represented as distributions or as labels is not a cosmetic difference; it determines whether the phenomenon can be observed at all. A selected-label analysis of these same data would have recorded 38.7% of human and 41.4% of model comparisons as showing no change when the distribution demonstrably moved (under the alphabetical tie-break convention; crediting the label method for tracking the complete set of tied top emotions instead lowers these to 25.8% for humans and 38.7% for models): for models, more than twice the comparisons it would have detected, and for humans nearly three-quarters as many as it would have detected. Even more directly, the same label-based treatment finds *no* significant human framing effect under the alphabetical tie-break convention used as the primary analysis, and one of six under the reverse-alphabetical convention also tested, where the distributional treatment finds three of six. Any study asking whether some manipulation shifts subjective interpretation is exposed to this, and how large the blind spot is only becomes clear once it is measured.

A second methodological result reaches just as far. Distributional distance between small vote samples is not zero when there is no effect, and it scales with the dispersion of the underlying distribution. Any ambiguity-defined grouping compares distances across groups that differ systematically in dispersion, which confounds the effect of interest with the measurement’s own noise floor. An explicit per-item sampling null makes this visible, and in this study it changed what could be concluded from H3.

7.3 Closing Remarks

This thesis set out to challenge a widespread practice: collapsing multiple human judgments into one gold label and treating the residue as noise [Basile 2020; Aroyo and Welty 2015]. The practice is convenient and entrenched, but for a substantial fraction of texts it is simply not well defined. Among the sampled texts (a sample stratified towards ambiguous categories, not a corpus-wide rate), one text in five has no emotion commanding a majority of readers, and 13.3% have no unique most-popular answer at all. For the texts with tied top choices, a pipeline reporting a single label depends on a tie-break; for texts with a unique plurality but no majority, that label still conceals the disagreement.

The distributional view shows that the disagreement being thrown away is not simply noise. Its overall magnitude and direction shift detectably when readers are told something about who wrote the text, in humans and models alike, and the two share that shift weakly but measurably. The claim concerns the aggregate shift, not every permutation pattern or category ordering examined along the way; several of those remain inconclusive at this sample size. Emotion recognition systems evaluated only against majority labels are scored on a target that discards variation this thesis has shown to be both structured and responsive to context.

Should models reproduce human interpretive uncertainty, or is it enough that they give human-like answers? That is a design question, not an empirical one. It cannot even be asked, though, while the evaluation target is a single label, and the results reported here suggest that humans and models differ in ways that matter.

Bibliography

- Acheampong, Francisca Adoma, Henry Nunoo-Mensah, and Wenyu Chen [2021]. “Transformer Models for Text-Based Emotion Detection: A Review of BERT-Based Approaches”. In: *Artificial Intelligence Review* 54, pp. 5789–5829. DOI: [10.1007/s10462-021-09958-2](https://doi.org/10.1007/s10462-021-09958-2). URL: <https://link.springer.com/article/10.1007/s10462-021-09958-2>.
- Aroyo, Lora and Chris Welty [2015]. “Truth Is a Lie: Crowd Truth and the Seven Myths of Human Annotation”. In: *AI Magazine* 36.1, pp. 15–24. DOI: [10.1609/aimag.v36i1.2564](https://doi.org/10.1609/aimag.v36i1.2564). URL: <https://doi.org/10.1609/aimag.v36i1.2564>.
- Basile, Valerio [2020]. “It’s the End of the Gold Standard as we Know it. On the Impact of Pre-aggregation on the Evaluation of Highly Subjective Tasks”. In: *AIxIA 2020 Discussion Papers Workshop*. Vol. 2776. CEUR Workshop Proceedings, pp. 31–40. URL: <https://ceur-ws.org/Vol-2776/paper-4.pdf>.
- Benjamini, Yoav and Yosef Hochberg [1995]. “Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing”. In: *Journal of the Royal Statistical Society: Series B* 57.1, pp. 289–300. DOI: [10.1111/j.2517-6161.1995.tb02031.x](https://doi.org/10.1111/j.2517-6161.1995.tb02031.x).
- Chou, Huang-Cheng and Chi-Chun Lee [2019]. “Every Rating Matters: Joint Learning of Subjective Labels and Individual Annotators for Speech Emotion Classification”. In: *ICASSP 2019 - IEEE International Conference on Acoustics, Speech and Signal Processing*, pp. 5886–5890. DOI: [10.1109/ICASSP.2019.8682170](https://doi.org/10.1109/ICASSP.2019.8682170). URL: <https://doi.org/10.1109/ICASSP.2019.8682170>.
- Davani, Aida Mostafazadeh, Mark Díaz, and Vinodkumar Prabhakaran [2022]. “Dealing with Disagreements: Looking Beyond the Majority Vote in Subjective Annotations”. In: *Transactions of the Association for Computational Linguistics* 10, pp. 92–110. DOI: [10.1162/tac1_a_00449](https://doi.org/10.1162/tac1_a_00449). URL: <https://aclanthology.org/2022.tacl-1.6/>.
- Demszky, Dorottya et al. [2020]. “GoEmotions: A Dataset of Fine-Grained Emotions”. In: *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Online: Association for Computational Linguistics, pp. 4040–4054. DOI:

- [10.18653/v1/2020.acl-main.372](https://aclanthology.org/2020.acl-main.372/). URL: <https://aclanthology.org/2020.acl-main.372/>.
- Fleiss, Joseph L. [1971]. “Measuring Nominal Scale Agreement Among Many Raters”. In: *Psychological Bulletin* 76.5, pp. 378–382. DOI: [10.1037/h0031619](https://doi.org/10.1037/h0031619).
- Geva, Mor, Yoav Goldberg, and Jonathan Berant [2019]. “Are We Modeling the Task or the Annotator? An Investigation of Annotator Bias in Natural Language Understanding Datasets”. In: *Proceedings of EMNLP-IJCNLP*, pp. 1161–1166. DOI: [10.18653/v1/D19-1107](https://doi.org/10.18653/v1/D19-1107). URL: <https://aclanthology.org/D19-1107/>.
- Good, Phillip I. [2005]. *Permutation, Parametric, and Bootstrap Tests of Hypotheses*. 3rd. Springer Series in Statistics. New York: Springer. ISBN: 0-387-20279-X.
- Henestrosa, Angelica Lermann, Hannah Greving, and Joachim Kimmerle [2023]. “Automated Journalism: The Effects of AI Authorship and Evaluative Information on the Perception of a Science Journalism Article”. In: *Computers in Human Behavior* 138, p. 107445. DOI: [10.1016/j.chb.2022.107445](https://doi.org/10.1016/j.chb.2022.107445).
- Holm, Sture [1979]. “A Simple Sequentially Rejective Multiple Test Procedure”. In: *Scandinavian Journal of Statistics* 6.2, pp. 65–70.
- Hong, Xin et al. [2025]. “AER-LLM: Ambiguity-aware Emotion Recognition Leveraging Large Language Models”. In: *ICASSP 2025 – 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, pp. 1–5. DOI: [10.1109/ICASSP49660.2025.10888198](https://doi.org/10.1109/ICASSP49660.2025.10888198).
- Pavlick, Ellie and Tom Kwiatkowski [2019]. “Inherent Disagreements in Human Textual Inferences”. In: *Transactions of the Association for Computational Linguistics* 7, pp. 677–694. DOI: [10.1162/tacl_a_00293](https://doi.org/10.1162/tacl_a_00293). URL: <https://aclanthology.org/Q19-1043>.
- Prabhakaran, Vinodkumar, Aida Mostafazadeh Davani, and Mark Díaz [2021]. “On Releasing Annotator-Level Labels and Information in Datasets”. In: *Proceedings of the Joint 15th Linguistic Annotation Workshop (LAW) and 3rd Designing Meaning Representations (DMR) Workshop*, pp. 133–138. DOI: [10.18653/v1/2021.law-1.14](https://doi.org/10.18653/v1/2021.law-1.14). URL: <https://aclanthology.org/2021.law-1.14/>.
- Röttger, Paul et al. [2022]. “Two Contrasting Data Annotation Paradigms for Subjective NLP Tasks”. In: *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Seattle, United States: Association for Computational Linguistics, pp. 175–190. URL: <https://aclanthology.org/2022.naacl-main.13/>.

- Santurkar, Shibani et al. [2023]. “Whose Opinions Do Language Models Reflect?” In: *Proceedings of the 40th International Conference on Machine Learning*. Vol. 202. Proceedings of Machine Learning Research. PMLR, pp. 29971–30004. URL: <https://proceedings.mlr.press/v202/santurkar23a.html>.
- Sap, Maarten et al. [2019]. “The Risk of Racial Bias in Hate Speech Detection”. In: *Proceedings of ACL*, pp. 1668–1678. DOI: [10.18653/v1/P19-1163](https://doi.org/10.18653/v1/P19-1163). URL: <https://aclanthology.org/P19-1163>.
- Troiano, Enrica, Laura Oberländer, and Roman Klinger [2023]. “Dimensional Modeling of Emotions in Text with Appraisal Theories: Corpus Creation, Annotation Reliability, and Prediction”. In: *Computational Linguistics* 49.1, pp. 1–72. DOI: [10.1162/colia_00461](https://doi.org/10.1162/colia_00461). URL: <https://aclanthology.org/2023.cl-1.1/>.
- Uma, Alexandra et al. [2021]. “Learning from Disagreement: A Survey”. In: *Journal of Artificial Intelligence Research* 72, pp. 1385–1470. URL: <https://doi.org/10.1613/jair.1.12752>.
- Waseem, Zeerak [2016]. “Are You a Racist or Am I Seeing Things? Annotator Influence on Hate Speech Detection on Twitter”. In: *Proceedings of NLP and Computational Social Science*, pp. 138–142. DOI: [10.18653/v1/W16-5618](https://doi.org/10.18653/v1/W16-5618). URL: <https://aclanthology.org/W16-5618>.
- Zhang, Saizheng et al. [2018]. “Personalizing Dialogue Agents: I have a dog, do you have pets too?” In: *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 2204–2213. DOI: [10.18653/v1/P18-1205](https://doi.org/10.18653/v1/P18-1205). URL: <https://aclanthology.org/P18-1205/>.

Saeed, Umer

LAST NAME, FIRST NAME

2093296

STUDENT REGISTRATION NUMBER

A Study of Large Language Models' Emotion Classification in
Ambiguous Text with Perceived Authorship

THESIS TITLE



Consent Form for Third Party Viewing of an Unpublished Final Thesis in the University Archives of the University of Bamberg

I consent that my thesis be made available to the users of the University Archives in accordance with the regulations of the Bavarian Archive Act (Bay-ArchG) and the University Archives' Terms of Use. Furthermore, I authorise the usage of my thesis, including quotation in an academic paper with appropriate indication of the source.

☒ Yes

☐ No*

Bamberg

29.09.26

A handwritten signature in black ink, appearing to be 'U. Saeed', is written over a horizontal dashed line.

Place and Date, Signature

* If consent is not granted, the thesis will remain in the examination files for 5 years. At the end of these 5 years' time, the thesis will be transferred to the University of Bamberg Archives where a decision on permanent storage will be reached.

Declaration of Authorship

In accordance with § 9 Para. 12 APO, I hereby declare that I wrote the preceding master's thesis independently and did not use any sources or aids other than those indicated. Furthermore, I declare that the digital version corresponds without exception in content and wording to the printed copy of the master's thesis and that I am aware that this digital version may be subjected to a software-aided, anonymised plagiarism inspection.

Date: 29.09.26 Signature: 